

Efficiency, Equity, and Cost-Recovery Trade-Offs in Municipal Water Pricing

Casey J. Wichman

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About the Author

Casey Wichman is an associate professor in the School of Economics at the Georgia Institute of Technology and a university fellow at Resources for the Future. He is an applied microeconomist working on issues at the intersection of environmental and public economics. His research focuses on how people interact with the natural and built environment, and what that behavior reveals about the value of environmental amenities. Wichman's research spans water and energy demand management, climate change impacts and policy, urban transportation, public goods provision, and outdoor recreation.

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Casey J. Wichman[†]

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Abstract

Municipal water utilities choose rates to recover costs, encourage conservation, and reduce burdens on low-income customers, which may deviate from optimal two-part tariffs. Theory suggests that prices should equal marginal cost with fixed costs recovered via fixed fees or alternative tax revenues. Using rate structure and municipal finance data for more than 700 utilities, I show that prices are discounted severely for low levels of consumption within nonlinear rate structures, leading to suboptimal usage and budget deficits, particularly in poorer and smaller communities. Marginal-cost pricing corrects allocative inefficiencies, and equity and cost-recovery goals can be achieved through more progressive approaches to fixed costs, which are both highly regressive and a large share of total costs.

Key Words: water pricing; utility management; natural monopolies; two-part tariffs; nonlinear pricing; affordability; municipal finance.

JEL codes: D12, H23, L95, Q25

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[†]Wichman: Associate Professor, School of Economics, Georgia Institute of Technology, 221 Bobby Dodd Way, Atlanta, GA 30332. Email: wichman@gatech.edu; University Fellow, Resources for the Future.

1 Introduction

Municipal water and wastewater utilities are tasked with the operation of critical infrastructure for the well-being of their residents: delivering clean, safe, and reliable drinking water and returning it to natural water bodies without harming the environment. Due to political pressure, concerns about affordability, and the United Nations' declaration that access to water is a human right (UN General Assembly, 2010), water prices typically do not reflect the true cost of service (Hanemann, 1997; Timmins, 2002b; Olmstead, 2010). Failure to recover costs can prevent utilities from investing in infrastructure and maintaining service quality (McRae, 2015). Even in developed countries, poor fiscal management can result in catastrophic consequences, such as recent service failures in Jackson, MS,¹ or Flint, MI (Christensen et al., 2023). Efficiency suggests that prices should equal long-run marginal costs of provision, but utilities—largely operated by financially and politically constrained municipal governments—need to balance equity considerations with revenue needs and clear market signals.² In balancing these goals, utilities often adopt nonlinear rate structures that discount marginal prices for low levels of consumption (Olmstead et al., 2007; Szabo, 2015; Wang and Wolak, 2022; Allaire and Dinar, 2022), although these pricing decisions may generate allocative inefficiencies (Borenstein, 2012, 2016; Borenstein and Davis, 2012), require subsidization from alternative municipal revenue sources (Timmins, 2002a; Renzetti, 1999), and ineffectively redistribute income (Levinson and Silva, 2022; Wietelman et al., 2024).

I evaluate the extent to which prevailing water prices deviate from economically optimal rates, what drives those inefficiencies, and what implications mispriced water services have on social welfare and utility revenue. I first revisit classic theory on two-part pricing in the spirit of Coase (1946), but I refocus this framework on contemporary issues in municipal water and wastewater pricing. The framework demonstrates the canonical efficiency result that price should equal marginal cost. However, unlike recent research concerning increasing-block rate (IBR) pricing for private, regulated electric utilities (Levinson and Silva, 2022), municipal managers can relax the utility's budget constraint via intragovernmental transfers. In practice, utilities often run deficits (subsidizing water service from general fund revenue) or surpluses (funding other local public goods through water rates), so assuming that the zero-profit condition holds is restrictive. Relaxing this constraint allows municipalities to draw from alternative sources of revenue to recover fixed costs. Fixed household service fees for water and wastewater—often \$250–350 per year or approximately 40–50 percent of water expenditures—operate as lump-sum taxes on households, which is likely the most regressive form of revenue raised by local governments.³

¹See, for example, <https://www.splcenter.org/news/2023/06/28/timeline-jackson-mississippi-water-problems>. (accessed June 19, 2024).

²Approximately 150,000 US water systems in the United States provide water and wastewater services to 90 percent of the US population; the vast majority of these are publicly owned (<https://www.epa.gov/dwreginfo/information-about-public-water-systems>).

³Fixed access fees do not vary by income. Property and sales taxes, the two most likely sources of municipal revenue, are also regressive, but consumption and property values tend to increase, albeit less than proportionally, with income.

Water utilities that care about equity can set the volumetric rate equal to marginal cost and, to deal with redistribution and recoup fixed costs, they can set an individualized fixed monthly service fee that targets differences in income (e.g., by setting service fees proportional to property values) or rely on less regressive sources of municipal revenue.

Empirically, I explore whether water prices deviate from optimal two-part tariffs using more than a decade of annual data on utility rate structures and municipal finance information for more than 700 utilities in the Southeastern United States.⁴ I provide evidence that utilities fail to balance their budget exactly and, on average, price water below average cost. Using variation in operating expenditures over time and across utilities in similar hydrologic regions, I estimate marginal costs for water and wastewater services and use these estimates to evaluate how much utilities discount or mark up water rates, accounting for nonlinearities in the rate structure. The predominant rate structures are IBRs and uniform rates that include an allocation of water at zero marginal price (“uniform + free”), both of which discount volumetric prices severely for low levels of use, accounting for nearly half of the consumption distribution. The median price discount for utilities with IBRs in the first tier (0–3 kgal/month) is approximately 34 percent, with prices in higher tiers marked up by 5–27 percent above marginal cost. The median utility that provides an allocation of free water in its uniform rate structure discounts prices by 65 percent in the first tier, with approximately marginal-cost pricing in higher tiers. Utilities with purely uniform rates price water, on average, close to marginal cost.

Why do utilities subsidize a substantial portion of consumption through discounted marginal prices? Common explanations are that the managers are influenced by political pressure to keep prices low (Timmins, 2002b) and provide basic water service at low cost for low-income customers (Szabo, 2015), despite maintaining a lump-sum transfer instrument on the bill that can address equity and/or affordability concerns more directly and more effectively. I explore the factors associated with inefficient pricing by comparing water rates and markups across utilities in similar hydrologic regions. Several results emerge. First, volumetric subsidies are much more common in smaller utilities, which have higher average costs of service; their inability to capitalize on economies of scale reduces the likelihood of passing the full costs through to customers. Second, volumetric prices are lower and more heavily discounted for utilities with larger low-income populations, which is notable because it reveals an efficiency–equity trade-off: these utilities are more likely to subsidize volumetric prices rather than alter the fixed service fee or cover fixed costs via municipal revenue. Third, water rates and markups are larger for communities with newer homes, more renters, and a higher proportion of non-White residents. Fourth, greater historical drought intensity increases volumetric prices but not markups, indicating that scarcity is potentially priced into water rates via higher marginal costs. Furthermore, I show that

⁴My sample includes the vast majority of utilities in Georgia and North Carolina, which are the eighth and ninth most populous states, respectively. Combined, their population (~21M) would represent the fourth largest state, after California (~39M), Texas (~31M), and Florida (~23M), comprising 6–7 percent of the US population. In terms of economic output, North Carolina’s GDP in 2023 was \$626B, and Georgia’s was \$661; combined, they would be comparable to Indonesia, the Netherlands, Turkey, and Saudi Arabia in world economic output.

these pricing decisions are consequential for utility operations: decreases in fixed fees and in volumetric prices in the lowest tier increase the likelihood that a utility fails to cover its costs.

Next, I estimate revenue elasticities, which indirectly reveals price elasticities of demand, leveraging year-to-year variation in the utility-level panel data on revenues and rate changes. I adopt an instrumental-variable (IV) strategy to account for nonlinear rate structures and potentially endogenous rate changes using rate structure characteristics from neighboring utilities. The estimated revenue elasticities suggest that for every 10 percent increase in volumetric prices, revenues increase by approximately 4–5 percent, which implies price elasticity estimates of -0.5 to -0.6 . Using these elasticities and the estimated marginal costs, I simulate revenue and social welfare implications of adopting efficient pricing. Many utilities can increase revenue by pricing at marginal cost, which is largely a direct surplus transfer from consumers. In most cases, that surplus declines due to removal of the inframarginal volumetric subsidy implicit in the nonlinear rate structures. Whether consumption increases or decreases depends on the presumed model of price perception and the wedge between prices and marginal costs. Marginal prices are generally close to, or slightly larger than, marginal cost, which implies that consumption increases slightly if consumers respond to marginal price (Olmstead et al., 2007; Nataraj and Hanemann, 2011). For utilities with nonlinear rate structures, average volumetric prices tend to fall below marginal costs due to inframarginal discounts, so consumption decreases if consumers respond to average price (Ito, 2014; Wichman, 2014; Brent and Ward, 2019; Shaffer, 2020). Regardless of price perception, nonlinear rates generate allocative inefficiencies, which are alleviated by marginal-cost pricing. For the median utility, marginal-cost pricing eliminates deadweight loss and raises revenue, which almost offsets consumer surplus losses. Although this is purely an efficiency result, the revenue gains can be used to advance distributional goals, by either adopting lower (or progressive) fixed fees or leveraging alternative, less regressive sources of municipal revenue to recover fixed costs.

This paper makes several contributions to our understanding of inefficiencies in utility pricing. I generate new insights on natural monopoly pricing that balances equity–efficiency trade-offs and document widespread pricing inefficiencies in the municipal water sector. This paper adds to a long-standing line of inquiry on optimal pricing for municipal utility services, some of which is focused on the redistributive implications of nonlinear rates (e.g., Borenstein, 2012; Borenstein and Davis, 2012; Levinson and Silva, 2022; Borenstein and Bushnell, 2022). Most research on water pricing adopts modeling frameworks from energy pricing, but water utilities are distinct from energy utilities in several important ways. First, they typically jointly manage production, distribution, and treatment, which is atypical of energy utilities. Relatedly, because of the prohibitive costs of moving water long distances, water utilities are smaller and managed much more locally. Both of these factors limit their ability to benefit from scale economies.⁵ As a result, per capita fixed costs of water service (and, hence, fixed charges) are relatively large

⁵To emphasize this point, Georgia has 52 municipally owned electric utilities, 41 electric membership corporations, and one investor-owned utility that operates in 155 of its 159 counties (source: <https://psc.ga.gov/utilities/electric/>). By contrast, I analyze 382 municipal water utilities in Georgia that serve, on average, 5,000 customers.

and deserve more explicit attention in water pricing. Second, water utilities are overwhelmingly operated by municipal governments, so rates are regulated indirectly by voters and typically not by a utility commission.⁶ This structure allows utilities to tap into alternative sources of municipal revenue, which are less regressive than a fixed lump-sum tax to recover fixed costs. Third, water service provides many local public goods, such as water provision for fire protection and stormwater infrastructure for flood protection, that are funded through utility bills and can rationalize municipal subsidies (Beecher, 2020).

This paper also contributes to our understanding of nonlinear pricing for utility services. Several studies explore the effects of IBRs for electricity, natural gas, and water use at the household level (e.g., Reiss and White, 2005; Olmstead et al., 2007; Nataraj and Hanemann, 2011; Ito, 2014; Wichman, 2014; Szabo, 2015), but little work has focused on utility-wide consequences, especially for water and wastewater services.⁷ Nonlinear rates are being increasingly adopted for water, largely under the banner of achieving conservation and reducing burdens on low-income customers. This research has largely focused on IBRs, but I additionally highlight a common hidden inefficiency in utility pricing decisions: many utilities possess rate structures that embed an allocation of free consumption (on average, 1–3 kgal/month per customer). These nonlinear volumetric rate structures do not reflect costs, often require subsidization from other revenue sources, and can lead to suboptimal consumption by driving a wedge between (infra-)marginal prices and marginal costs. I show that uniform, marginal-cost pricing can eliminate allocative inefficiencies and increase revenue by removing volumetric subsidies, which could be used, in principle, to lower fixed fees or offset them with less regressive sources of municipal revenue.

Last, comparative analyses of water utilities are rare because no centralized repository of data links water rate structures with municipal revenue and cost information for the same utilities over time. Most research on municipal water pricing evaluates consumer responses to price and rate structure changes using billing data—typically focusing on one or a small set of utilities—with the goal of estimating price elasticities of demand (e.g., Olmstead et al., 2007; Olmstead, 2009; Mansur and Olmstead, 2012; Baerenklau et al., 2014; Wichman et al., 2016; El-Khattabi et al., 2021; Smith, 2022). In contrast, I characterize the efficiency costs of prevailing water pricing practices using a decade of water and wastewater rate structures matched with administrative data on municipal finances for more than 700 municipal utilities. To my knowledge, the data assembled in this paper is the most extensive panel data set of water utility rate structures linked with utility costs and revenues. These data allow me to estimate marginal costs and compare rate structure characteristics across utilities in varied geographic regions (i.e., coastal, mountainous, urban,

⁶Private or for-profit but not municipal water utilities are typically regulated by state utility commissions.

⁷Research has addressed the complications of IBRs because they generate kinks in consumer budget constraints (Hanemann, 1984). Structural discrete-continuous choice models have been applied to estimate price elasticities of demand for water and electricity (e.g., Hewitt and Hanemann, 1995; Olmstead et al., 2007; Olmstead, 2009; Baerenklau et al., 2014; Szabo, 2015; McRae, 2015). However, researchers have also questioned whether consumers understand nonlinear rate structures and provided evidence, for both electricity and water, that consumers might respond to average, rather than marginal, prices (Ito, 2014; Wichman, 2014; Brent and Ward, 2019; Shaffer, 2020). Others have argued that consumers do respond to marginal price if the changes are large enough (Nataraj and Hanemann, 2011).

rural) and with heterogeneous service populations. Importantly, the data set includes hundreds of small- and medium-sized water utilities that are rarely evaluated in economics research despite their central role in delivering essential services to residents outside of large metropolitan areas. Renzetti (1992), Renzetti (1999), and Timmins (2002b) are among the few papers that evaluate the efficiency consequences of municipal water (and wastewater) prices, but they are based on now-outdated data sets with limited geographic or temporal scope and do not account for the recent, increasing prevalence of nonlinear rate structures despite the trade-offs inherent in fixed versus volumetric pricing decisions.

2 Conceptual framework

In this section, I lay out a model of rate-setting for municipal water utilities. My approach adopts the general framework of Levinson and Silva (2022), which analyzes the redistributive potential of nonlinear electricity pricing by extending a formal model of Coase (1946), but I refocus it on the particulars of municipal water and wastewater utilities.

2.1 The household's problem

Consider a household that maximizes utility over water consumption, w^i , and a numeraire good x^i , where superscripts index households ($i = 1, \dots, n$). The household earns income M^i , which is subject to a marginal tax rate of $\tau \geq 0$ and a lump-sum government transfer g^i (both of which the household takes as given). The household spends income on water (at marginal price p with a fixed service fee $t \geq 0$) and the numeraire good (with price normalized to 1) in the present period, leading to the budget constraint $pw^i + t + x = (1 - \tau)M^i + g^i$. Let $\hat{M}^i \equiv (1 - \tau)M^i + g^i - t^i$ represent after-tax residual income. The household's problem becomes

$$\max_{w_i} u^i(w_i, \hat{M}^i - pw^i). \quad (1)$$

The household's first-order condition is

$$\frac{u_w^i}{u_x^i} = p \text{ for all } i = 1, \dots, n. \quad (2)$$

This condition implicitly defines demand for water and the numeraire good: $w^i(p, \hat{M}^i)$ and $x^i(p, \hat{M}^i) = \hat{M}^i - pw^i(p, \hat{M}^i)$. Plugging these demand equations into the utility function results in the following indirect utility function: $v^i(p, \hat{M}^i) = u(w^i(p, \hat{M}^i), x^i(p, \hat{M}^i))$ for all $i = 1, \dots, n$.

2.2 The government's problem

The government chooses prices, fixed fees, and marginal tax rates to maximize the sum of indirect utility households receive. Its role is the first point of departure from Levinson and Silva (2022)

because the majority of US water utilities are public entities owned and operated by municipal governments; they are largely not private, regulated natural monopolies, such as natural gas or electric utilities. This distinction has important implications for budget balancing and fixed-cost recovery that I explore directly.

The government manages a water utility that has fixed costs F and per-unit costs c . So, the government's total cost of supplying $W = \sum_i w^i$ units of water is $F + c \sum_i w^i$. The government must balance its budget each year, resulting in the following budget constraint:

$$\sum_i \left[t + pw^i + \tau M^i \right] = F + c \sum_i w^i + G, \quad (3)$$

where $G = \sum_i g_i$ are total governmental transfers. The left-hand side of this equation is governmental revenue; the right-hand side is governmental expenditure.

2.2.1 A self-sufficient water utility

First, assume the government chooses p and t but treats τ as fixed and exogenous and that $\tau \sum_i M^i = G$, so that it focuses only on balancing revenue and costs. This results in the simpler budget constraint $\sum_i [t + pw^i - cw^i] = F$, which reduces to the baseline case in Levinson and Silva (2022). The solution, outlined in detail in Appendix A, provides the canonical result that price should be set equal to marginal cost ($p = c$). Substituting this result into the government's budget constraint results in setting fixed fees equal to average fixed costs, or, $t = F/n$.

If the government can instead set individualized prices, p_i —which might reflect price discrimination across customers or heterogeneous marginal prices across the consumption distribution (i.e., IBRs)—the primary insight does not change. The optimal pricing rule when individualized prices are feasible results in the condition that $p_i = c$ for all i , or that all units of consumption should be charged the same marginal price. This result—that marginal price should be equal across customers and uses—should be uncontroversial to economists and suggests that any pricing structure that deviates from marginal-cost pricing will be inefficient. Borenstein (2012) claims that IBRs for electricity have no cost basis, as IBRs assign different marginal prices to different customers, leading to different incentives to consume water.⁸

Levinson and Silva (2022) additionally show that if the government cares about equity and can set individualized fixed service fees and marginal prices, then marginal price should still be set to equal to marginal cost, for all households and uses, and the fee used to redistribute income completely (i.e., if households differ only in their incomes, then it will be used to equalize income across all households). Of course, fully redistributing income through a utility bill is an impractical result, although milder versions of this solution may aid in redistribution without sacrificing efficiency.⁹ Levinson and Silva use this as a point of departure from optimal two-part tariffs to

⁸Hanemann (1997), however, argues that IBRs could be supported by a cost argument, inasmuch as block rate structures reflect the shape of increasing marginal costs in the long run.

⁹For example, one could restrict service fees to be nonnegative and fix their levels to be proportional to household income or property values, which has been proposed previously (e.g., Borenstein et al., 2021; Burger et al., 2020;

consider how IBRs might be used as a second-best solution to achieve distributional goals when the regulator cannot use individualized access fees or other revenue-raising instruments.

2.2.2 Allowing government transfers

Municipal water utilities can be distinguished from private, regulated energy utilities because the municipality can set local taxes to raise revenue and cross-subsidize the water utility through intragovernmental transfers. I now reintroduce the government's capacity to raise revenue through income taxes.¹⁰ In addition to choosing prices and fixed fees, the government also chooses a tax rate τ to balance its budget. Its problem is now given by

$$\max_{p, t, \tau} \sum_i v^i(p, \hat{M}^i) + \mu \left[\sum_i [t + \tau M_i + p w^i - c w^i - g_i] - F \right] \text{ for } i = 1, \dots, n, \quad (4)$$

where $\hat{M}^i = (1 - \tau)M^i + g^i - t$ and μ reflects the Lagrangian multiplier on the balanced budget constraint. The necessary equilibrium conditions are

$$[\text{FOC for } p] : \frac{\partial v^i}{\partial p} + \mu[w^i + (p - c)w_p^i] = 0 \quad (5)$$

$$[\text{FOC for } t] : -\frac{\partial v^i}{\partial \hat{M}} + \mu[1 - (p - c)w_M^i] = 0 \quad (6)$$

$$[\text{FOC for } \tau] : -\frac{\partial v^i}{\partial \hat{M}} + \mu[1 - (p - c)w_M^i] = 0 \quad (7)$$

The first-order conditions for t and τ are identical. Following the same argument, taking the ratio of Equation 5 and Equation 6 (or 7), and solving for the resulting optimal marginal prices provides the identical solution that $p = c$ for all households and uses.

Because of the equality in the equilibrium conditions for t and τ , however, neither revenue-raising instrument has a unique solution to cover fixed costs or public goods. To see this, substitute $p = c$ into the budget constraint and solve for the fixed fee as a function of τ :

$$t = \frac{F + G - \tau \sum_i M^i}{n}, \quad (8)$$

which demonstrates that the fixed fee should be set equal to average fixed costs plus average government transfers less average revenue raised via income taxes. Considering the case where $G = 0$, it is easy to see that t will be smaller than in the case where the utility must balance its own budget for any $\tau > 0$. Thus, any deviation from marginal-cost pricing (i.e., $p = c$) is suboptimal.

Wietelman et al., 2024). This approach would not be optimal but would reallocate costs in a comparatively progressive way relative to spreading fixed costs evenly across all customers.

¹⁰Municipal governments may have a variety of income, property, and sales taxes that they can control, but I focus on an income tax to maintain simplicity, to allow household heterogeneity to be summarized by income, and because income taxes are likely good proxies for alternative wealth and consumption taxes.

But, absent other distortions, the government can choose any combination of $t \geq 0$ and $\tau \geq 0$ that satisfies Equation 8 to cover fixed costs and public expenditures. So, unlike in other settings (e.g., for a private, regulated electric utility), municipal utilities that have multiple instruments to raise revenue need not set fixed access fees equal to average fixed costs, which results in a departure from Coasean two-part tariffs. One implication is that the government might subsidize the utility from its general tax revenue or the utility might cross-subsidize alternative public expenditures.

In practice, many utilities adopt a form of nonlinear rate structures to address distributional goals by subsidizing low levels of consumption through discounted inframarginal prices (e.g., Szabo (2015); McRae (2015); Levinson and Silva (2022)).¹¹ Other utilities adopt “lifeline” rates that reduce the volumetric price of water for low-income customers. Those policies are misguided if (i) they lead to deviations from marginal-cost pricing or (ii) less regressive sources of revenue are available to cover fixed costs. If a utility has preferences over equitable outcomes but charges a common t to all households, this fee acts as a flat lump-sum tax and is likely the most regressive form of municipal revenue. Water bills are regressive: Cardoso and Wichman (2022) show that the lowest decile of US households spends approximately 7 percent of their income on essential water use, whereas median-income households spend approximately 1 percent. To make this comparison more concrete, I illustrate the regressivity of property and income taxes relative to water and wastewater expenditures using data from the 2016 Consumer Expenditure Survey in Figure 1. It represents the declining share of income allocated to each type of expenditure as a proportion of the share that the lowest income quintile pays. As shown, water bills are more regressive than property or sales taxes for respondents in the states in my sample (North Carolina and Georgia). As income increases, households spend a proportionally smaller share on water services than alternative sources of governmental revenue. Households in the third income quintile pay about 23 percent of the share of income on sales tax that first-quintile households pay. For water, this same statistic is about 16 percent. Focusing solely on the fixed charge for water services (effectively an annual \$300 lump-sum tax) is even more regressive.

Water utilities can achieve efficiency goals via marginal-cost pricing and address distributional goals through either an individualized fixed service fee—perhaps set proportionally to property values, which is information readily available to municipal governments—or alternative, more progressive, sources of municipal revenue (e.g., income, property, or sales taxes). In the model, I have ignored any distortionary costs of raising revenue through alternative instruments, so the government is indifferent between using t or τ to balance its budget. Increases in property, income, or sales taxes that arise from loading fixed costs of water service onto the tax base may distort housing and labor markets, although these costs would be a relatively small share of property and sales tax revenue and be borne more heavily by wealthier households. Additionally, as I show empirically later on, many utilities are already subsidizing volumetric water

¹¹Timmins (2002a) explores a version of this question in a median voter framework, where voters prefer to scale marginal water prices by their share of income, resulting in cross-subsidization of the water utility via lower marginal prices. However, the focus of that model is not on two-part tariffs, so the role of the fixed service fee is not considered. This is an important distinction because fixed fees comprise a substantial portion of the typical water bill.

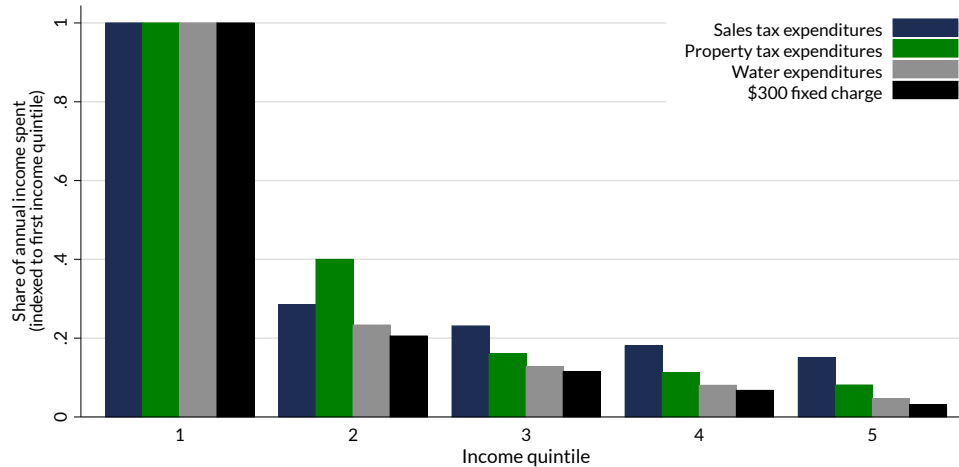


Figure 1: Share of income spent on alternative expenditures by income quintile

Notes: Data from the 2016 Consumer Expenditure Survey (CEX) for respondents in North Carolina and Georgia. The chart represents share of income spent on each category normalized to that for households in the first decile for comparison across different baseline levels. Sales tax expenditures, not reported in the CEX, are calculated simply as 7 percent (for North Carolina) and 8 percent (for Georgia) of total annual expenditures (less expenditures on housing, health, and education). Property tax expenditures and “water and other public services” are reported in the CEX. Expenditures on an annual \$300 fixed charge for water service is presented for comparison. Statistics are weighted for nationally representative sampling design.

prices with general fund revenue, so the aggregate change in welfare would be ambiguous.

2.3 Which marginal cost?

The previous subsection suggests that prices should be set equal to marginal cost. Many economists argue that water rates should reflect the long-run marginal cost (LRMC) of provision (e.g., Renzetti, 1992; Hanemann, 1997; Olmstead, 2010), reflecting service costs both today and in the future. Because water service is particularly capital intensive, and the cheapest supplies are developed first, LRMCs are likely increasing. Hanemann (1997) notes that this may provide some justification for adopting IBRs to reflect the shape of increasing long-run costs.¹²

Pricing at LRMC requires utilities to know the incremental costs of provision now, in addition to the cost of developing new water supplies in the future. This undertaking is likely infeasible in many dimensions, especially given variability in projected water supplies as a result of climate change and variability in future demand due to population growth and changes in preferences and technology. Utilities are more likely to know, or be able to estimate, the short-run marginal cost (SRMC). Water can be purchased on the wholesale market, but utilities predominantly extract it freely and bear the costs of treatment and distribution that vary with quantity supplied. As a result, a utility’s operating costs can be used to measure SRMC. It may be even easier for utilities to price water at its average total cost of service, although average costs may be larger or smaller than (long-run) marginal costs, leading to incorrect incentives for water use.

If water consumption generates externalities, private LRMCs may still be too low. For wa-

¹²However, rather than doing so, utilities often adopt IBRs as a means to either encourage conservation among large-volume users or redistribute costs across the consumption distribution.

ter utilities, externalities could be crowding out alternative uses of scarce water for neighboring municipalities, agricultural uses, or ecosystem services (Olmstead, 2010; Ayres et al., 2021). Treatment and distribution of water and wastewater is highly energy intensive, which generates social costs via air pollution and greenhouse-gas emissions from electricity generation. Additionally, social costs may arise from poor water quality, from failure to either provide clean drinking water (e.g., Christensen et al., 2023; Keiser et al., 2023) or account for downstream pollution from insufficient wastewater treatment (e.g., Keiser and Shapiro, 2019; Keiser et al., 2019).¹³ Because of these unpriced social costs, the social marginal cost likely exceeds LRMC (and, hence, SRMC).

2.4 Measuring inefficiencies

I demonstrate illustrative welfare changes from allocative inefficiencies that arise from inefficient pricing for a representative water consumer in Figure 2. I plot a typical two-tier IBR structure relative to two different marginal costs.¹⁴ In the left panel, the IBR structure has two blocks in which both marginal prices fall below marginal cost (MC_H). The demand curve $W(p)$ intersects with the second block, p_1 , resulting in consumption, w_0 . Because both marginal prices are smaller than marginal cost, too much water is consumed, resulting in deadweight loss equal to area H . Setting prices at MC_H would eliminate deadweight loss by reducing the quantity of water demanded to w_H^{MC} . This switch would unambiguously decrease consumer surplus by $B + C + E + F$ (i.e., the area below MC_H and above $p(w)$ and bounded on the right by $W(p)$). Some of this consumer welfare loss is captured as a transfer to the government as new revenue (area $B + C$) because previously discounted water is now priced at marginal cost. Because quantity demanded decreases, however, the government also loses revenue from area G . Whether revenue increases or decreases on net depends on the relative magnitudes of the transfer from consumers and the demand response to marginal-cost pricing. Notably, if $p(w)$ is a uniform rate, with all units charged at p_1 , then marginal-cost pricing eliminates deadweight loss and results in unambiguous consumer surplus losses and municipal revenue gains so long as demand is inelastic.

Now consider the right panel of Figure 2, which shows the same rate structure and demand curve but smaller marginal costs, MC_L . Marginal costs fall between the two marginal prices ($p_0 < MC_L < p_1$). This example is, perhaps, more akin to how IBRs are adopted in practice because utilities often seek to maintain revenue neutrality within IBRs by subsidizing consumption on the lower tier through increased revenue at higher consumption levels. Demand intersects $p(w)$ on the second tier, which results in lower than optimal quantity demanded (w_0) relative to marginal-cost pricing (w_L^{MC}) and deadweight loss equal to area F . Marginal-cost pricing would eliminate this deadweight loss and increase consumption, resulting in an ambiguous consumer surplus

¹³Although water quality externalities are not directly related to the quantity supplied, the costs of compliance with such regulations are incorporated into the cost of water and wastewater services (Jerch, 2022; Meyer and Raff, 2024).

¹⁴A more formal welfare analysis of consumer demand under nonlinear pricing might adopt a discrete-continuous choice framework to deal with the complications that arise from kinked budget constraints (e.g., Burtless and Hausman, 1978; Hanemann, 1984; Olmstead, 2009; Szabo, 2015). I abstract from that discussion, as my focus is not on consumer choice in particular but rather highlighting the aggregate social costs of inefficient pricing, changes in utility revenue, and the implicit transfers between consumers and municipalities.

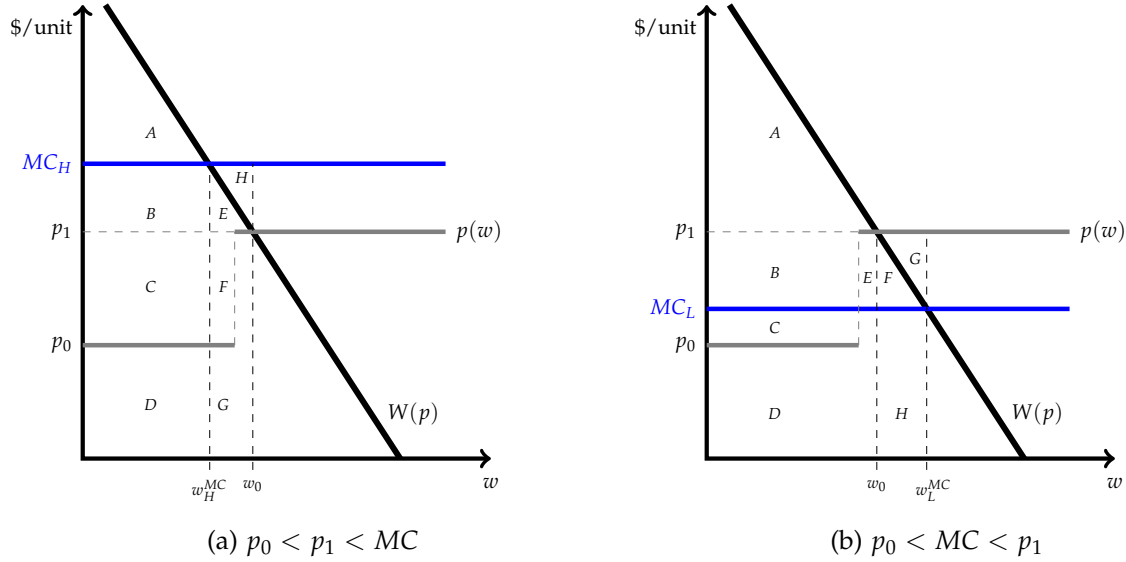


Figure 2: Illustrative welfare framework for nonlinear pricing

Notes: **Panel (a):** The prevailing rate structure is $p(w)$, with efficient price MC_H . Initial consumption (w_0) exceeds efficient consumption (w_H^{MC}), resulting in deadweight loss of area H . Setting price equal to MC_H results in unambiguous consumer surplus loss equal to $B + C + E + F$. Municipal revenue changes by area $B + C - G$, where $B + C$ represents a transfer from consumers to municipal water providers. **Panel (b):** The prevailing rate structure is $p(w)$, with efficient price MC_L . Initial consumption (w_0) is smaller than efficient consumption (w_L^{MC}), resulting in deadweight loss of area F . Setting price equal to MC_L results in a change in consumer surplus equal to $E + F - C$. Municipal revenue changes by area $C + H - E$, where $C - E$ represents a transfer from consumers to municipal water providers.

change of $E + F - C$, which accounts for the gains from the demand response ($E + F$) less losses from paying a higher inframarginal price (C). The inframarginal welfare loss is a transfer to municipal revenue, resulting in an ambiguous welfare revenue change of $C + H - E$. If $p(w)$ is a uniform rate, with all units charged at p_1 , then consumer surplus strictly increases and municipal revenue strictly decreases from marginal-cost pricing.

This framework demonstrates that the welfare effects of marginal-cost pricing depend critically on the levels of prevailing marginal prices relative to marginal cost and the sensitivity of demand to price changes. With uniform rates, welfare effects follow textbook examples, as described in Borenstein (2016). With nonlinear rates, however, welfare and revenue effects are ambiguous. Moreover, this analysis illustrates the implicit subsidy that nonlinear rates provide to consumers. A primary goal of the following empirical analysis is to quantify the divergence of volumetric prices and marginal costs at different levels of consumption and to estimate demand elasticities, which permit calculation of welfare effects of inefficient municipal pricing.

3 Data and empirical setting

3.1 Water and wastewater rate structures

Water and wastewater rates data are obtained from UNC-Chapel Hill's Environmental Finance Center (EFC). Each year, the EFC conducts a rate survey in which utility representatives supply

information about these rate structures and other utility characteristics. I focus on rate surveys from Georgia and North Carolina because these states have strong coverage over time, allowing for constructing utility-level panel data on rate structures for 732 utilities.¹⁵ For Georgia, rates data are available 2008–2019. For North Carolina, rates data are available 2010–2019. These data include information on billed amounts for water and wastewater service at different levels of consumption, type of rate structure, water source, and number of customer accounts each year. For observations that are missing the number of accounts, I linearly interpolate this value within each utility. Water and wastewater are charged on separate rate structures and can be operated by distinct entities; I combine these rates into an aggregate rate structure if both are present within the municipality. Approximately 80 percent of utilities in the sample provide combined water and wastewater service, and the remainder provides water service only.

From this data set, I construct several variables. Billed amounts for different levels of consumption (0, 3, 5, 10, and 15 kgal per month) are given. From these billed amounts, I construct average marginal prices (*AMP*) between each consumption threshold: $AMP_{3kgal} = [\text{Bill}_{3kgal} - \text{Bill}_{0kgal}] / 3$, $AMP_{5kgal} = [\text{Bill}_{5kgal} - \text{Bill}_{3kgal}] / 2$, $AMP_{10kgal} = [\text{Bill}_{10kgal} - \text{Bill}_{5kgal}] / 5$, and $AMP_{15kgal} = [\text{Bill}_{15kgal} - \text{Bill}_{10kgal}] / 5$. These variables capture nonlinear rate structures by approximating the degree to which marginal prices change at different levels of consumption. For uniform rates, average marginal prices are constant across consumption thresholds.

3.2 Municipal finance data

Water utilities are typically operated by towns or cities, although they may be operated by counties or consolidated city-county governments. I obtain municipal finance data for each year for these levels of municipal government. For Georgia, this information is from the Local Government Financial Documents database.¹⁶ For North Carolina, Annual Financial Information Reports are publicly accessible from the state treasurer’s office. From these databases, I gather annual operating costs and expenditures for each municipal utility, among other variables. These revenue data are combined for both water and wastewater utilities if both exist and are operated by the municipality.¹⁷ These data are matched with the rates data via fuzzy matching on utility name with manual inspection and correction.

These aggregate data do not provide detailed breakdowns of utility finances. I do not observe capital expenditures or debt service for utilities. My focus is on comparing average and marginal costs with fixed and volumetric prices, although I do not observe fixed costs in the data. Operating expenditures are more likely to reflect variable costs, as they capture the costs of treatment, water purchases, energy, and labor for operations and maintenance, all of which vary

¹⁵I was able to match 966 utilities in the rates data with municipal finance data, although I impose a series of sample restrictions and data quality checks, which trims the sample but results in a more credible and consistent sample for analysis. Wastewater-only utilities were removed along with utilities with fewer than eight annual observations, stemming from missing rates data. I report the sample restrictions and data quality checks in detail in Appendix B.1.

¹⁶Source: <https://ted.cviog.uga.edu/financial-documents/>.

¹⁷Several municipalities have water utilities but no wastewater utilities. The converse is also true (e.g., if the region has a private water provider), but this is rare, and I remove wastewater-only utilities from my data set.

with quantity of water supplied. Long-run costs would also include the large capital investments of building new infrastructure (e.g., a new reservoir or wastewater treatment facility). Additionally, debt service is a cost utilities must pay regardless of output. As a result, relying solely on operating expenditures to measure utility costs is a lower bound on private costs of water service provision but a good approximation of variable costs.

Neither the water rates data nor the municipal finance sources report reliable data on total quantity of water supplied.¹⁸ Instead, I recover implied quantity supplied for each utility-year from total revenue and the residential rate structure. Total revenue is a function of fixed monthly fees times the number of customers plus quantity consumed times the average volumetric price. I use this information to back out the implied quantity supplied for the typical household in each utility for each year.¹⁹ This process is described in detail in Appendix B.

3.3 Socioeconomic, geographic, and climatic data

For each municipality/county, I collect annual economic and demographic data from the Census ACS 5-year survey. These data provide information on household income, household size, race, and so forth. I match the municipality/county name from the water rates data to Census “place” names and county names using a fuzzy merge with manual correction. For 40 utilities, I cannot assign a direct match by name, so I use socioeconomic characteristics from their county. Utility boundaries and municipal/county boundaries do not align perfectly, so these variables are measured with error but provide a reasonable approximation of community characteristics.²⁰ I also gather information on Republican vote share in presidential elections at the county level from the MIT Election Lab.

For each utility, I determine the “hydrological unit” in which the municipality/county centroid falls. I obtain shapefiles for these units from the US Hydrological Survey’s Watershed Boundary Dataset and spatially match them with municipality/county centroids. I use HUC6 regions, which measures riversheds and captures common shocks to the hydrological system. The sample has 21 unique HUC6 regions. To incorporate measures of contemporaneous and historical water scarcity, I calculate the number of weeks each year that a given county was in severe or extreme drought as reported by the USGS Drought Monitor. If a county has any of its area in “Severe Drought,” I count that entire week as a week in severe drought. Severe drought is

¹⁸The municipal finance data for North Carolina do report quantities but were unreliable, likely resulting from inconsistent reporting and translation of units.

¹⁹This process results in implied monthly mean consumption per customer for each utility in my sample of 7.2 kgal/month, with a median of 5.7 kgal/month and 10th–90th percentile range of 2.1–14.8 (see Table 1). Summary statistics broken down by state in Appendix Table D.1 show means of 8.6 kgal/month and 5.7 kgal/month for Georgia and North Carolina, respectively. For context, Wichman et al. (2016) report average monthly consumption of around 5 kgal per month for a sample of 1,727 households in six municipal utilities in North Carolina, with the 10th and 90th percentiles at 2 and 9 kgal/month. Ferraro and Price (2013) report average monthly usage of roughly 8.5–9 kgal/month for summer 2007 in Cobb County, GA, directly preceding a behavioral experiment. Wichman and Ferraro (2017) additionally report virtually identical levels of monthly usage for two neighboring counties in the Atlanta, GA, metropolitan region.

²⁰For 2008–2010 for municipalities in Georgia and 2010 in North Carolina, I use the 5-year ACS from 2011 due to data availability. For 2008 for counties in Georgia, I use the 5-year ACS data from 2009 due to data availability.

the central of five categories reported in the Drought Monitor data (the more extreme categories are "Extreme" and "Exceptional"). For each county, I construct annual measures of drought and a rolling, cumulative exposure variable equal to the annual average number of weeks in each category for in the preceding 10 years. I aggregate PRISM daily maximum and minimum temperatures and precipitation levels to the county-by-year level (Schlenker and Roberts, 2009).

3.4 Descriptive statistics

Summary statistics are presented Table 1 for the first annual observation for each utility. The average utility provides a half-million gallons of water each year and serves approximately 5,800 customers, with substantial variation in both variables. The 10th percentile utility serves only 225 customers; the 90th percentile utility serves 9,502. The sample also captures very large utilities, including in the Atlanta, GA, metropolitan region (i.e., City of Atlanta, Cobb County, DeKalb County, Gwinnett County, City of Gainesville); Charlotte, NC; Greensboro, NC; Raleigh, NC; and Winston-Salem, NC; all of which have more than 100,000 residential customers. I present a “map” (a scatterplot of utility locations by latitude and longitude) weighted by number of customers in Figure D.1 to illustrate the heterogeneity in utility size. Mean (median) monthly usage per customer is 7.2 kgal (5.7 kgal). Approximately half of utilities have groundwater supplies, with the remainder split equally between surface water and purchased water.²¹

Monthly bills for 5, 10, and 15 kgal per month average \$51.6, \$88.7, and \$126.7. Rates are generally higher in North Carolina, and consumption is generally larger in Georgia. These total billed amounts reflect fixed service fees plus volumetric rates. The average fixed service fee is \$25, indicating that for typical levels of consumption (5 kgal/month), the fee makes up almost 50 percent of the monthly bill. Average marginal prices of 0–3, 3–5, 5–10, and 10–15 kgal per month are \$4.1, \$7.1, \$7.4, and \$7.6, reflecting the nonlinearity of the typical rate structure with lower marginal prices for lower consumption. These averages mask substantial variation across utilities: for the 0–3 kgal/month tier, the 10th–90th range of marginal prices is \$0.0–\$9.2 per kgal. Average bills (evaluated at 5 kgal/month) are \$10.3 per kgal. Average costs of supply (defined by total annual expenditures divided by total quantity supplied) are even larger, averaging \$13.1 per kgal.

In Section 2.2.2, I suggested that municipal utilities may violate the zero-profit assumption via cross-subsidization. Here, I provide empirical support for that claim. I present operating ratios—defined as operating revenue divided by operating costs—in Table 1. On average, this ratio is 1.1, suggesting that the average utility covers its operating costs. Ratios below 1 indicate the utility brought in insufficient revenue to match its expenditures that year. There is variation across utilities, with 10th–90th percentile ranges spanning 0.8–1.4. To probe this variation further, I plot

²¹Information on water source is inconsistently reported in the rates data for some utilities, and, for others, the source changes over time. I use the most common source reported for each utility. Some utilities report multiple sources (e.g., groundwater and purchased in some years and purchased only in other years), and I list the most common. In rare cases that have no information on water source, I assume that it is groundwater, which is the most common.

Table 1: Summary statistics

	Mean	SD	10th %ile	50th %ile	90th %ile	N
Annual water supplied (million gallons)	0.5	1.9	0.0	0.1	1.0	732
Number of customers	5,776.5	20,190.8	225.0	1,284.5	9,502.0	732
Avg. monthly usage (kgal/month/customer)	7.2	5.2	2.1	5.7	14.8	732
Operating ratio (revenue/expenditures)	1.1	0.4	0.8	1.0	1.4	732
Combined water/wastewater utilities (%)	79.6	40.3	0.0	100.0	100.0	732
Source: Groundwater (%)	52.3	50.0	0.0	100.0	100.0	732
Source: Surface water (%)	23.4	42.3	0.0	0.0	100.0	732
Source: Purchased water (%)	24.3	42.9	0.0	0.0	100.0	732
Total bill: 5 kgal (\$)	51.6	21.6	26.0	50.1	78.7	732
Total bill: 10 kgal (\$)	88.7	40.2	41.7	84.2	137.9	732
Total bill: 15 kgal (\$)	126.9	61.1	57.1	119.4	202.1	732
Fixed service fee (\$)	25.1	12.5	11.5	23.8	41.2	732
Base free water allocation (kgal/month)	1.5	1.2	0.0	2.0	3.0	732
Base free sewer allocation (kgal/month)	1.2	1.3	0.0	1.0	3.0	583
Avg. marginal price: 0–3 kgal (\$/kgal)	4.1	3.6	0.0	3.2	9.2	732
Avg. marginal price: 3–5 kgal (\$/kgal)	7.1	3.9	2.7	6.4	12.0	732
Avg. marginal price: 5–10 kgal (\$/kgal)	7.4	4.1	2.9	6.7	12.6	732
Avg. marginal price: 10–15 kgal (\$/kgal)	7.6	4.4	2.7	6.8	13.2	732
Avg. total bill: 5 kgal (\$/kgal)	10.3	4.3	5.2	10.0	15.7	732
Avg. total cost (\$/kgal)	13.1	9.8	4.2	10.3	27.4	732
Median household income (\$ 1,000s)	37.5	13.5	22.6	35.1	53.8	732
Median home value (\$ 1,000s)	127.9	85.7	65.8	108.6	195.3	732
Pct. owner occupied	62.7	13.1	47.0	62.4	80.1	732
Prop. White (%)	64.3	21.7	34.3	66.0	92.8	732
Prop. Black (%)	29.4	21.5	2.7	26.2	60.4	732
Prop. Hispanic (%)	6.6	8.0	0.0	4.1	15.5	732
No. weeks with severe drought, 2010-2019	66.4	51.8	9.0	61.0	134.0	732

Notes: All observations are at the utility level for first year present in sample. For Georgia, data begin in 2008; for North Carolina, data begin in 2010. Number of utilities increases over time. All values are adjusted to 2019 dollars. Annual number of customers served is linearly interpolated and extrapolated within a utility if values are missing.

operating ratios for all utilities in 2015 in Figure 3. This figure shows dispersion in operating ratios, suggesting that many utilities generate either too little or too much revenue relative to their operating costs. The trend is increasing across utility size. On average, larger utilities are more likely to cover their operating costs. Deficits are more strongly concentrated among smaller utilities that may not benefit from economies of scale, whereas larger utilities more commonly experience surpluses that could be used to cover capital expenditures, debt service, or subsidies to other municipal services. By the same token, however, operating ratios of exactly (or slightly greater than) 1 indicate that utilities are *only* covering operating expenditures, which would suggest insufficient revenue for capital costs and debt service.

3.5 Rate structures

I present summary statistics for utility rate structures at the beginning and end of the sample in Table 2. I focus primarily on (a) uniform rates, in which all units of water are priced at the same volumetric price; (b) uniform rates that include a free allocation (i.e., “uniform + free”), which are

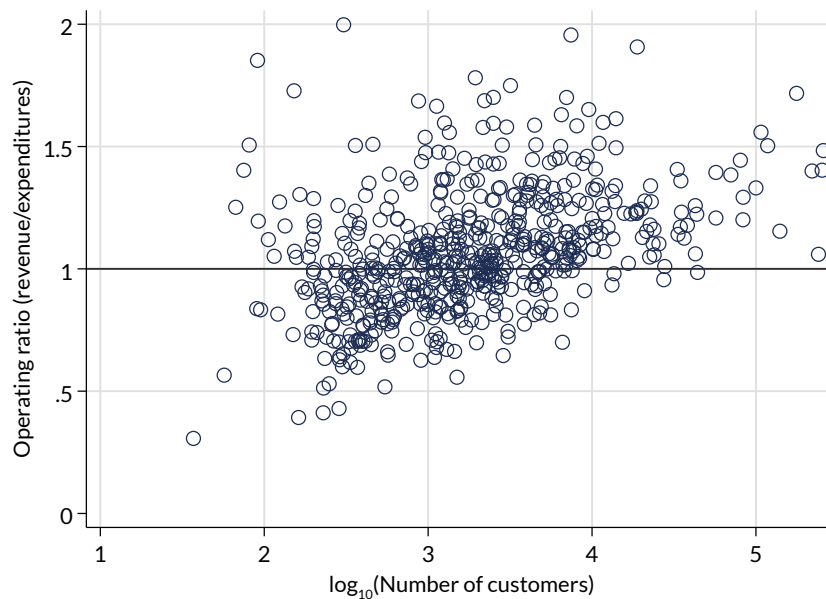


Figure 3: Annual operating ratios by size of utility

Notes: Figure shows annual operating revenues divided by operating expenditures conditional on size of the utility. Ratio is truncated (from above) at 2. Data presented are for 2015. Ratios greater than 1 indicate utility revenues exceed costs.

essentially IBRs with two tiers and a marginal price of zero in the first tier;²² (c) IBRs, which have an effective increase in the marginal price of water as consumption increases; and (d) all other rates, which includes fixed flat monthly fees, decreasing block rates, and block rates that blend increasing and decreasing segments. At the beginning of the sample, 17 percent of water rates are uniform, 39 percent of rates are “uniform + free,” 32 percent are IBRs, and 12 percent are other. By the end of the sample, 13 percent of water rates are uniform, 38 percent are “uniform + free,” 43 percent are IBRs, and 5 percent are other. This indicates a shift toward IBRs, which is consistent with trends documented in California (Allaire and Dinar, 2022), although California has even greater penetration rates of IBRs. For wastewater, the pattern is similar: penetration rates for IBRs increase from 21 to 29 percent throughout the sample, “uniform + free” increase from 41 to 44 percent, and uniform rates decrease from 31 to 23 percent. Although water and wastewater costs are separated on the bill, wastewater consumption is unmetered, which means that volumetric usage is inherently tied to water usage. As a result, it makes sense to analyze the implied joint rate structure, which tracks the penetration rates for water rate structures closely.

3.6 Costs, bills, and prices

To put the theoretical discussion into context with the data, I present average costs, average bills, and average marginal prices by each decile of water supplied in Figure 5. For utilities in the

²²“Uniform + free” rates are reported as uniform rates, which masks important nonlinearities inherent in seemingly simple rate structures. I categorized uniform rates into “uniform + free” if the rate structure reports that any positive amount of consumption is included at no volumetric price. In Appendix Figure D.2, I show that the majority of all rate structures include some level of free consumption, with the modal allotment being 2 kgal/month.

Table 2: Rate structures in use at beginning and end of sample

(a) First rate structure observed for each utility								
Water rates			Wastewater rates			Combined rates		
	N	%		N	%		N	%
IBR	233	31.8	IBR	123	21.1	IBR	241	32.9
Uniform	124	16.9	Uniform	184	31.6	Uniform	140	19.1
Uniform + free	285	38.9	Uniform + free	236	40.5	Uniform + free	305	41.7
Other	90	12.3	Other	40	6.9	Other	46	6.3

(b) Last rate structure observed for each utility								
Water rates			Wastewater rates			Combined rates		
	N	%		N	%		N	%
IBR	318	43.4	IBR	169	28.5	IBR	324	44.3
Uniform	99	13.5	Uniform	135	22.7	Uniform	101	13.8
Uniform + free	276	37.7	Uniform + free	264	44.4	Uniform + free	283	38.7
Other	39	5.3	Other	26	4.4	Other	24	3.3

Notes: "IBR" indicates increasing-block rates, "uniform" indicates uniform rates, "uniform + free" indicates a uniform rate with some allocation of water provided at zero marginal price, and "other" indicates all other rate structures (e.g., decreasing-block rates, flat charges). Panel (a) shows first observed rate structure for each utility (732 water and wastewater utilities and 594 water-only utilities). Panel (b) shows the last observed rate structure for each utility. "Combined rates" represents the implied rate structure from combining water and wastewater rates if both are present in the municipality. If either water or sewer rates are "uniform + free," then the combined rate structure is "uniform + free"; if either water or sewer rates are "IBR," then the combined rate structure is "IBR," which can supersede "uniform + free."

lowest decile, average costs are substantial: \$15–\$28 per kgal for the lowest five deciles. This trend continues to decline until it flattens out for the 8th–10th decile. This pattern reflects the water industries' large fixed costs that can be spread over an increasingly large customer base. Mean average bills (evaluated at 5 kgal/month) for each decile are presented in blue diamonds and remain largely flat across the supply distribution. This trend suggests that bills do not necessarily reflect costs. Average bills fall below average costs for utilities in the bottom half of the supply distribution, are comparable to average costs for the 5th–6th decile, and exceed costs for the 7th–10th decile. This comparison suggests that prices do not reflect average costs for smaller utilities and that utilities may not be generating sufficient revenue to cover costs. As 5 kgal/month is slightly smaller than average consumption per household-month, these average bills are likely larger than the average prices experienced by the typical consumer in each utility.

I also present average marginal prices for the lowest two tiers (0–3 and 3–5 kgal/month). Marginal prices in the lowest tier reflect nonlinear rates in which the first several gallons of consumption are discounted. On average, the marginal price for the first tier falls below average cost at all deciles. The average utility in the first decile pays more than \$25 to deliver the first 3 kgal per month but recovers only ~\$6 per kgal in return. The pattern is similar for the second tier, which generally reflects the marginal price of water that the average customer experiences. This value, however, approaches average costs only for the utilities in the 8th–10th decile of supply. These trends begin to reveal the degree of inefficiency in water rates. In what follows, I unpack these trends to understand whether prices might better reflect marginal costs.

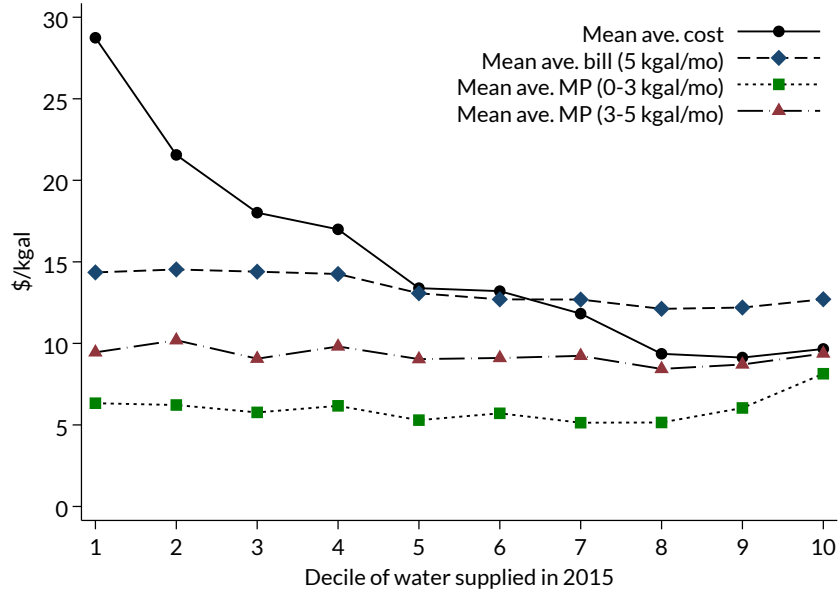


Figure 4: Average costs, average bills, and marginal prices by decile of water supplied

Notes: Average cost is total cost divided by implied quantity supplied. Data presented are for 2015 for utilities that provide both water and wastewater service.

4 Empirical strategy and results

In this section, I describe each facet of the empirical approach followed immediately by a discussion of the results. The goal of this analysis is to provide evidence of utility pricing decisions, particularly how they deviate from optimal two-part tariffs. I first implement an empirical test of average cost pricing. I then estimate marginal costs using cross-sectional variation in utility costs and explore the extent to which prices reflect marginal costs, identifying the prevailing inefficiencies in water rates. Next, I explore what community and utility characteristics are associated with inefficient pricing. Last, I estimate revenue elasticities that facilitate an analysis of welfare and revenue changes if utilities adopted marginal-cost pricing.

4.1 Average cost pricing

In Figure 5, I present the relationship between average billed amounts and average total costs. This figure shows a scatterplot reflecting the correlation between average bills at 5 kgal/month and average costs. The correlation is positive: as average costs increase, average billed amounts increase. The correlation is not perfect, however, perhaps demonstrating the unwillingness of utilities to fully pass through high costs. The solid black line weights the relationship by number of customers; the dashed red line weights each utility equally. Weighting by customers generally increases the correlation between average bills and average costs, suggesting that larger utilities are more likely to price closer to average cost, but the slope is less than 1. To test whether utilities price water at average costs more formally, I regress average total bills at 5 kgal per

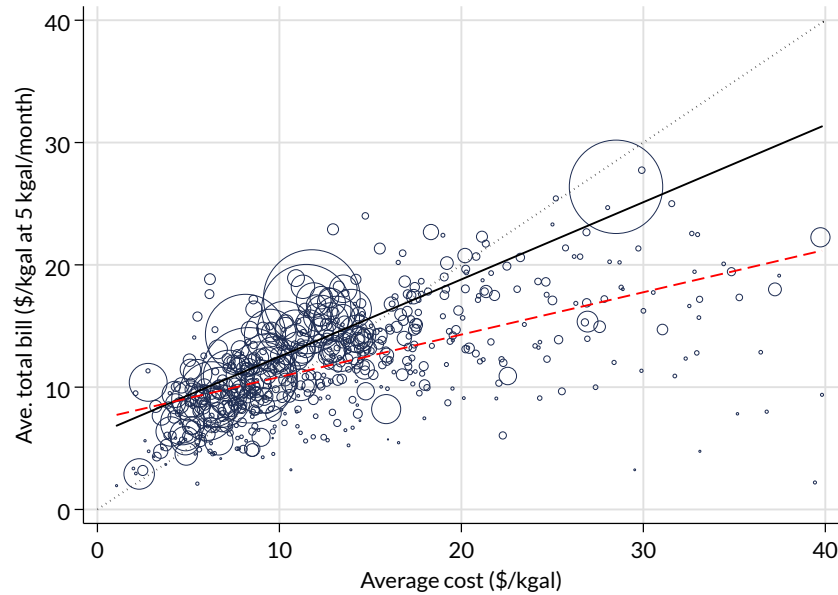


Figure 5: Average total bill (evaluated at 5 kgal/month) vs. average total cost

Notes: Figure shows average total bill (evaluated at 5 kgal/month) relative to average total cost. Both variables are in \$ per kgal. Size of circle indicates number of accounts serviced by each utility. Black (solid) line represents linear regression weighted by number of accounts. Red (dashed) line represents linear regression unweighted. Dotted line is 45-degree line. Data presented are for 2015. Observations with average total cost greater than \$40/kgal are truncated.

month on average annual costs, including year-by-HUC6 fixed effects, weighting by the number of residential customers in each utility. The coefficient estimates, presented in Table C.1, are small, ranging from 0.62 in Georgia to 0.41 in North Carolina, and averaging 0.53 for the full sample. These estimates suggest that for every dollar that costs increase, utilities pass through \$0.53 on the typical water bill at 5 kgal/month. There is heterogeneity across utility size: utilities with fewer than 1,000 customers pass through approximately one-quarter of changes in cost to customers (26 cents on the dollar). This same statistic is 33 percent for medium (5,000–10,000 customers) and 70 percent for large (>10,000 customers) utilities. A coefficient of 1 would suggest that, on average, utilities price water with respect to average cost. I present additional results—including unweighted and lagged regressions—that support this basic result and further discussion in Appendix C.

4.2 Estimating marginal costs

If utilities are not pricing water at average cost, perhaps it is close to marginal cost, which could be larger or smaller. Data on marginal costs for water and wastewater services are not readily available—especially on the East Coast, where water inputs are largely self-supplied—so I adopt an approach that estimates utility-specific marginal costs from aggregate data on operating costs. In the municipal finance data, I observe expenditures for utilities that supply different amounts of water each year. The cross-sectional relationship of aggregate costs reveals how incremental

costs of water supply vary as quantity increases. I specify the following equation:

$$\ln(\text{Expenditures}_{iht}) = \alpha + \beta_1 \ln(Q_{iht}) + \beta_2 \ln(Q_{iht}) \times 1[\text{Wat. only}_i] + \lambda_{ht} + \varepsilon_{iht}, \quad (9)$$

where $\text{Expenditures}_{iht}$ represents annual utility operating expenditures for utility i in hydrological region h in year t , Q_{iht} is implied annual quantity supplied, and λ_{ht} is a HUC6-by-year fixed effect reflecting annual unobserved variation within the six-digit hydrological region. I include an interaction of $\ln(Q_{iht})$ with $1[\text{Wat. only}_i]$ indicating that the utility provides only water service (and not wastewater service). I estimate Equation 9 separately for each of the three water sources observed in the data (groundwater, surface water, and purchased water), as these may have different cost structures associated with extraction or purchase. I also substitute λ_{ht} for λ_{htq} , where q represents utility size groups (i.e., <5,000, 5,000–10,000, or >10,000 customers). This additional specification leverages variation within a hydrological region each year for utilities of roughly similar size that have common input costs, such as local labor costs, or variation in geology (e.g., depth to groundwater).

Equation 9 estimates the relationship between costs and expenditures and is akin to a log-linear cost function. Because I do not observe input prices or cost shares, this approach relies on the assumption that they are constant within each hydrological region each year and absorbed by the λ_{ht} (or λ_{htq}) fixed effect. This assumption is reasonable given that many costs (e.g., labor, energy) are not likely to vary much within small geographic regions. By differentiating Equation 9, similar to Koetter et al. (2012) and Delis et al. (2014), utility-specific marginal costs by year can be recovered directly from estimates of Equation 9 and observed data, which amounts to scaling a utility's average cost by the estimated coefficient $\hat{\beta}_1$. That is, $MC_{it} = (\text{Expenditures}_{iht} / Q_{iht}) \hat{\beta}_1$. For water-only utilities, $\hat{\beta}_1 + \hat{\beta}_2$ is substituted in for $\hat{\beta}_1$ to account for the interaction term.

These estimates can be interpreted as something between a short- and long-run marginal cost. The use of contemporaneous operating expenditures and quantities suggests that $\hat{\beta}_1$ and $\hat{\beta}_2$ would reflect short-run costs, although because they are estimated using cross-sectional variation across utilities within the same hydrological region, this estimate additionally reflects changes in incremental costs as utilities supply more water. As smaller utilities grow, comparisons to larger utilities that annually supply more water reveals the incremental costs utilities can expect to bear as a result of economies of scale and development of new water supplies.^{23,24}

I present coefficient estimates of Equation 9 in Table 4. The first three columns include HUC6-by-year fixed effects, and the final three columns add an additional level of fixed effects that

²³As mentioned, data on capital expenditures are not available, although capital investment is an important component of understanding the true costs utilities face in the long run as they develop new water sources and build/replace infrastructure and treatment facilities. Accordingly, marginal cost estimates from Equation 9 should be viewed as a lower bound to LRMC. Moreover, focusing on operating expenditures only, rather than capital expenditures, is more likely to capture variable costs, which understates the true costs of water service.

²⁴Not all utilities grow in size. As Davis and Hausman (2022) show, shrinking populations in service areas for natural gas utilities can cause utilities to move “backward” along the LRMC curve, resulting in fewer customers to spread fixed costs over. This has no bearing on the estimation of marginal costs in Equation 9 but does have policy importance for understanding the true costs utilities face as demographics change.

Table 4: Marginal cost regression estimates

	(1)	(2)	(3)	(4)	(5)	(6)
	Surface	Ground	Purchased	Surface	Ground	Purchased
$\ln(Q)$	0.81*** (0.03)	0.78*** (0.01)	0.81*** (0.02)	0.76*** (0.04)	0.62*** (0.02)	0.69*** (0.03)
$\ln(Q) \times 1[\text{Wat. only}]$	-0.04*** (0.01)	-0.07*** (0.01)	-0.05*** (0.01)	-0.04*** (0.01)	-0.08*** (0.01)	-0.05*** (0.01)
Observations	1594	3192	1514	1551	3097	1450
No. Utilities	170	383	176	167	375	170
R-squared	0.93	0.90	0.93	0.94	0.92	0.95
State FE?	Y	Y	Y	Y	Y	Y
HUC6-by-Year FE?	Y	Y	Y	—	—	—
HUC6-by-Year-by-Size FE?	—	—	—	Y	Y	Y

Notes: The dependent variable is logged annual operating expenditures at the utility level. “1[Wat. only]” indicates utilities that only provide water service. The first column estimates a constant marginal cost across all utilities. Regressions are estimated separately for each water source (Surface, Groundwater, and Purchased). HUC6-by-Year-by-Size fixed effects refers to six-digit hydrological region-by-year-by-utility size, where utility size captures three categories of utility size by number of customers: 0–1,000, 1,000–5,000, and >5,000. Standard errors are clustered at the utility level. Note that sample sizes change because some observations are dropped due to limited variation within HUC6-year(-by-size) for each water-source group.

account for utility size groups. Each column is a separate regression that isolates heterogeneity in water source. These coefficients can be interpreted directly as the percentage of total expenditures that vary with output: a coefficient of 0.7 indicates that as utilities supply 10 percent more water each year, their operating expenditures increase by 7 percent, or that 70 percent of total operating expenditures are variable costs. The preferred estimates in Columns (4)–(6) provide coefficients of 0.76 for surface water, 0.62 for groundwater, and 0.69 for purchased water. Utilities that provide water only have slightly smaller coefficients.

Figure 6 illustrates the distribution of utility-specific marginal costs for 2015, calculated as $MC_{it} = (\text{Expenditures}_{iht} / Q_{iht})\hat{\beta}_1$, as described previously. I plot marginal costs alongside average costs. Marginal costs are slightly smaller than average costs, indicating the relative proportion of fixed and variable costs that are embedded in operating expenditures. As a robustness check, I replicate this same framework with a more flexible trans-log-like cost function approach in Table D.2 and Figure D.3. The results are virtually identical.

4.3 Comparing prices to marginal costs

How do utilities’ prices compare to these estimated marginal costs? First, I provide a graphical illustration of prices relative to marginal costs for each type of rate structure in Figure 7 for a central year of the sample: 2015. These box plots present the distribution of marginal prices at different levels of consumption, capturing nonlinearities in the rate structure, relative to the estimated marginal costs (where the horizontal red line reflects the median marginal cost estimate). In Panel (a), for combined water–wastewater utilities, utilities with uniform rates, on average, price water approximately at marginal cost. However, for uniform rates that include a free allocation of water, the first tier of consumption is priced well below marginal cost, with the

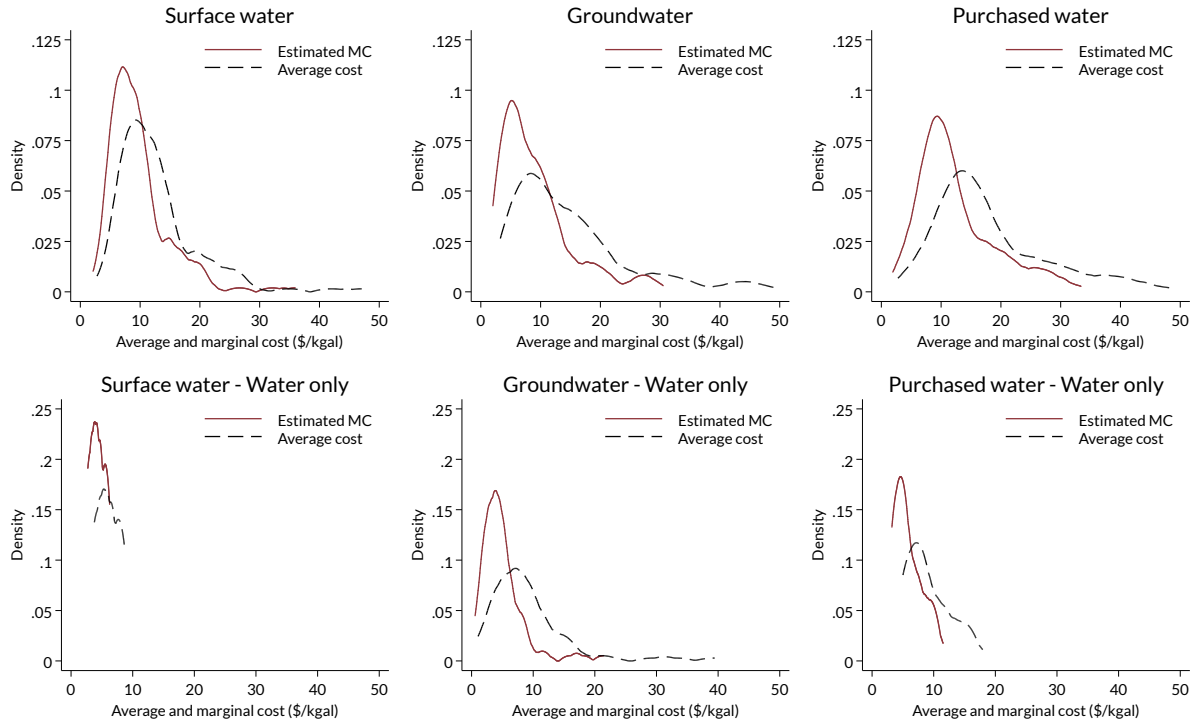


Figure 6: Average and estimated marginal costs

Notes: Marginal costs are estimated from Columns (4)–(6) in Table 4 with HUC6-by-year-by-utility size fixed effects. Estimated coefficients from the log-linear cost function in Equation 9 scale average costs. Data presented are for 2015.

remainder of consumption priced closer to it. For IBRs, similarly, the first tier of consumption is priced below marginal cost, the second tier is priced closer to it, and the third and fourth tiers are priced above it. In Panel (b), I present the same results for water-only utilities. These results generally follow the same pattern as combined water-wastewater utilities. The variation across utilities in both panels includes substantial heterogeneity.

In Table 5, I present more concrete statistics for marginal price markups at different levels of consumption for 2015. Markup is defined as $(Avg. MP - MC)/MC$.²⁵ I continue to break out markup by rate structure, but I pool estimates for combined utilities and water-only utilities. For uniform rates, markups are effectively zero for the mean and median utility. The distribution of utilities has considerable variation, with the 10th–90th percentile ranging from ~44 percent discounts to ~55 percent markups. For uniform rates with a free allocation, the first tier is discounted 66 percent (68 percent) relative to marginal costs for the mean (median) utility. For higher tiers, however, prices generally reflect marginal costs. Notably, the utility in the 10th percentile discounts the first tier of water use by 100 percent. For IBRs, markup increases in each tier. For the mean (median) utility, the first tier is discounted by 26 percent (34 percent), the second tier is marked up by 10 percent (5 percent), and the third and fourth tiers are marked up

²⁵Note that this definition of markup is simply percentage change in prices relative to marginal cost, rather than the more conventional $(p - MC)/p$. I adopt this definition because it is generally more intuitive and because observed marginal prices can be zero, particularly for low levels of consumption.

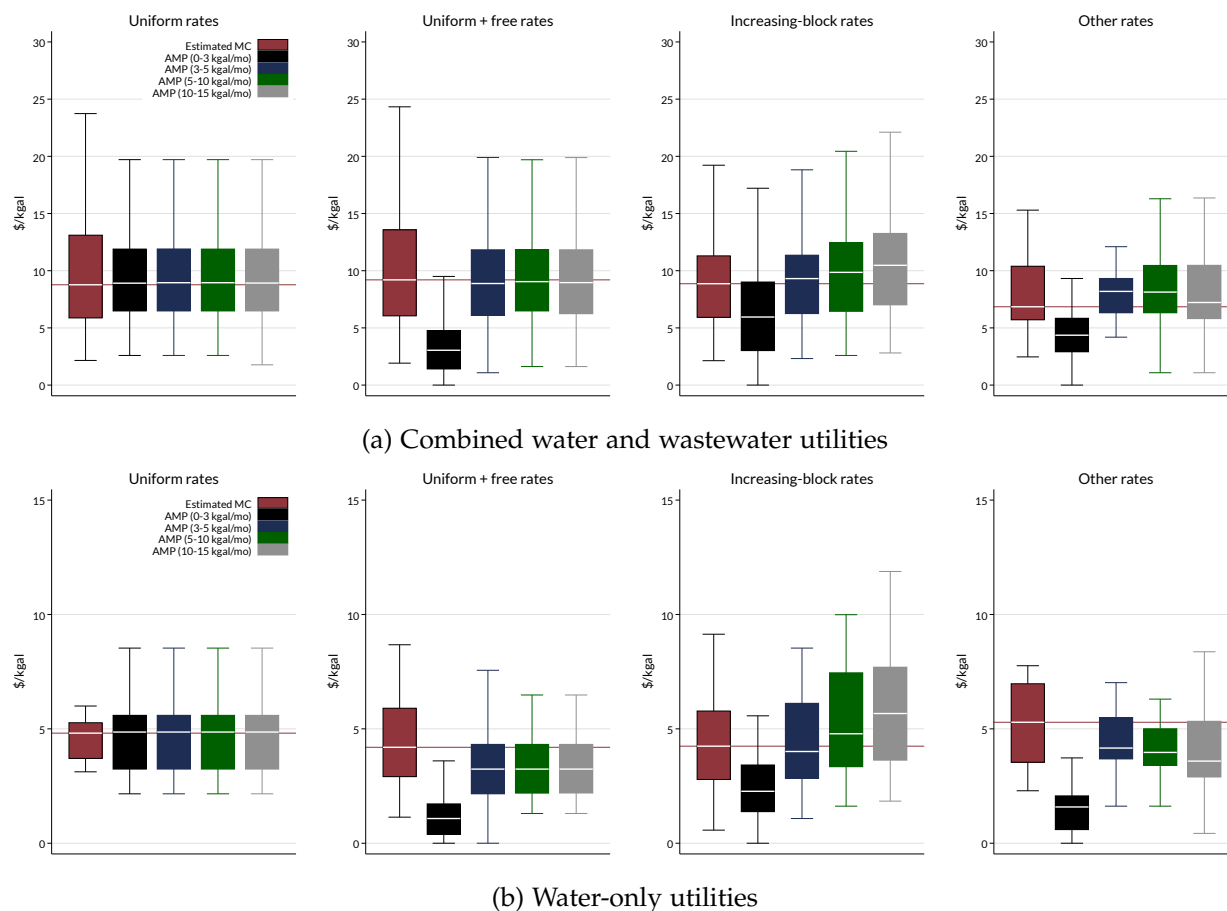


Figure 7: Average marginal prices relative to marginal cost

Notes: Plots represent 2015 data only. Central horizontal line is 50th percentile. Height of box represents 25th–75th percentiles. Whiskers represent 1.5 times the interquartile range. Horizontal red lines represent the median of estimated marginal costs. Uniform rates include rate structures that provide an allocation of consumption at zero marginal price.

by 22 percent (17 percent) and 34 percent (27 percent). For other rate structures, the first tier is discounted substantially, but higher tiers are similar to marginal cost.²⁶

Overall, Figure 7 and Table 5 tell a consistent story. Utilities with uniform rates generally price water close to marginal cost. Utilities with uniform rates that include a free allocation discount costs dramatically, although the higher tiers are priced approximately at marginal cost. IBRs, similarly, discount the first tier of consumption, but higher tiers are priced above marginal cost. These findings are consistent with the design of, and nonlinearities embedded within, each rate structure, but the comparison to marginal cost reveals the implicit volumetric subsidy that utilities provide to low levels of consumption. Because the median utility sells approximately 6

²⁶The statistics in Table 5 reflect one central year of the sample: 2015. In Figure D.4, I show mean markup by water source for each year, 2010–2019. I exclude data from 2008 and 2009 because water rates data for North Carolina begin in 2010. This figure suggests that discounts and markups have remained relatively stable, with a slight increase over time. All marginal prices and expenditures are adjusted to 2019 dollars, so these trends reflect increases in markups on top of inflation. These trends are not necessarily surprising: water rates rarely move downward, with a documented trend that water and wastewater rates have been rising faster than inflation for several decades (Cardoso and Wichman, 2022).

Table 5: Markup results for all utilities by rate structure

Panel A: Uniform rates					
	Mean	SD	10th %ile	50th %ile	90th %ile
Mark-up of Avg. MP at 0-3 kgal/mo	0.05	0.39	-0.41	0.01	0.47
Mark-up of Avg. MP at 3-5 kgal/mo	0.06	0.39	-0.41	0.03	0.48
Mark-up of Avg. MP at 5-10 kgal/mo	0.06	0.39	-0.41	0.03	0.48
Mark-up of Avg. MP at 10-15 kgal/mo	0.05	0.40	-0.45	0.03	0.48
Panel : Uniform + free rates					
	Mean	SD	10th %ile	50th %ile	90th %ile
Mark-up of Avg. MP at 0-3 kgal/mo	-0.65	0.29	-1.00	-0.68	-0.30
Mark-up of Avg. MP at 3-5 kgal/mo	-0.01	0.50	-0.60	-0.00	0.53
Mark-up of Avg. MP at 5-10 kgal/mo	0.01	0.50	-0.58	-0.00	0.58
Mark-up of Avg. MP at 10-15 kgal/mo	0.00	0.50	-0.58	-0.00	0.58
Panel B: Increasing-block rates					
	Mean	SD	10th %ile	50th %ile	90th %ile
Mark-up of Avg. MP at 0-3 kgal/mo	-0.26	0.51	-0.87	-0.34	0.44
Mark-up of Avg. MP at 3-5 kgal/mo	0.10	0.48	-0.47	0.05	0.70
Mark-up of Avg. MP at 5-10 kgal/mo	0.22	0.53	-0.41	0.16	0.87
Mark-up of Avg. MP at 10-15 kgal/mo	0.34	0.66	-0.38	0.27	1.05
Panel C: Other rates					
	Mean	SD	10th %ile	50th %ile	90th %ile
Mark-up of Avg. MP at 0-3 kgal/mo	-0.47	0.49	-1.00	-0.65	0.18
Mark-up of Avg. MP at 3-5 kgal/mo	0.05	0.53	-0.56	-0.02	0.68
Mark-up of Avg. MP at 5-10 kgal/mo	0.01	0.50	-0.56	-0.07	0.68
Mark-up of Avg. MP at 10-15 kgal/mo	-0.01	0.55	-0.68	-0.02	0.68

Notes: Markup is defined as $(Avg. MP_{it} - MC_{it}) / MC_{it}$, where MC_{it} are the utility-specific marginal cost estimates from Table 4, Columns (4)–(6). Each row represents average marginal price at different levels of monthly consumption. Avg. MP at 0–3 kgal/mo represents the average marginal price up to 3 kgal per month (not including the base service fee). Avg. MP at 3–5 kgal/mo represents the average marginal price for 3–5 kgal per month, and so forth. Markup results shown are for 2015.

kgal per month per customer, and because more than 75 percent of utilities have IBRs or uniform + free rate structures, these findings suggest that the vast majority of utilities discount nearly half of their volumetric water consumption by a substantial degree.

4.4 Understanding inefficient pricing

Next, I explore what community or utility factors are associated with inefficient rate-setting practices. I estimate a series of regressions that relate utility, socioeconomic, and environmental characteristics to changes in rates and markups. Separately, I regress the fixed service fee, average marginal prices at different levels of consumption, and price markups on utility size, community income, age of homes, proportion of households that are renter occupied, proportion of non-White residents, Republican vote share, the average number of drought weeks experienced in the previous 10 years, and indicators for water sources. I describe the rationale for each of these variables next. I include HUC6-by-year fixed effects, so identification comes from cross-sectional variation across utilities and within utilities over time, accounting for any common

HUC6-specific year effects (e.g., regional drought) and unobserved state effects (e.g., capturing differences in state regulatory policy).

The number of customers captures how rates correlate with increases in utility size. As it increases, utilities may pass through cost-savings from economies of scale. Second, nonlinear rates are often adopted to maintain equitable water bills while sending a conservation signal to high-use households. So, one might expect utilities with more low-income customers to have lower, marked-down volumetric prices. Furthermore, Section 2 suggests that equity goals can be achieved through fixed fees rather than marginal prices. To test whether socioeconomic characteristics are associated with different types of rate structures, I include regressors for the proportion of households with annual income $\leq \$50k$ or $> \$100k$. Age of infrastructure is another likely driver of utility costs; I do not observe infrastructure age, but I proxy for it using the proportion of homes built before 1980 (i.e., roughly capturing homes built before the Clean Water Act, which pushed utilities to build and upgrade wastewater treatment facilities during the 1970s (Keiser et al., 2023)) or after 2005 (i.e., indicating relatively new infrastructure).

Because water rates are set by local governments, political factors also play a role. I include the proportion of residents in the county who voted for the Republican presidential candidate in the 2012 election as a proxy for political conservatism. I also include the proportion of renter-occupied homes. Renters may not pay utilities directly and may have a smaller political voice, so it may be easier for municipalities with more renters to pass through a greater share of costs. I also include the proportion of the population that is non-White to evaluate whether rates are higher or lower in communities with greater minority populations.

Additionally, utilities may adopt rate structures in response to experience with drought. The Southeast had several periods of extreme drought since 2000.²⁷ Although water scarcity is generally not priced into municipal water rates, utilities may adjust rates after a drought to encourage conservation or recoup lost revenue from nonprice conservation policies implemented during a drought. To capture these effects, I define a variable equal to the average number of weeks per year that at least some portion of the county is classified as “Extreme Drought” (i.e., the second-highest classification of the US Drought Monitor) over the previous 10 years. This variable is a rolling average and defined at the county level.

The results are presented in Table 6. I first focus on the number of customers, which illustrates how economies of scale are captured within rate structures. In the first column, I find that for every 10,000 additional customers, fixed service fees decrease by \$1.10. This result is intuitive: as utilities grow in size, they can spread their fixed costs across a broader customer base, which is consistent with why average costs fall across deciles of water supply in Figure 5. In Column (2), marginal prices in the lowest tier increase as the number of customers increase. Moreover, markups in all price tiers increase with utility size. The magnitude of these effects are modest, however, as a 10,000-customer increase is larger than moving from the 10th to 90th percentile in

²⁷Georgia experienced at least some counties in the highest category (“Exceptional Drought”) in 2000, 2002, 2007–2008, 2011–2013, and 2016; North Carolina had Exceptional Droughts in 2000, 2002, 2007–2008, and 2016.

the sample. Overall, larger utilities have higher fixed fees and higher marginal prices.

Next, I focus on the relationship between rates and income. In the first column, income has no relationship with base service fees, suggesting that utilities that serve to more low- or high-income populations do not use the fixed access fee to redistribute fixed costs. In Columns (2)–(5), however, I find evidence that utilities with proportionally larger low-income populations have lower marginal prices (\$0.03–0.4 decreases in marginal prices, or 0.66–1.21 percentage point markdowns) for every percentage point increase in the proportion of households with income below \$50k. Moving from the 25th to the 75th percentile of income would lower marginal prices by \$0.56/kgal, \$0.33/kgal, \$0.46/kgal, and \$0.69/kgal for each of the four marginal prices, respectively. These estimates suggest that utilities with more low-income households have lower marginal prices on average. As the number of high-income (>\$100k) households increase, marginal prices do not change, but markups increase slightly (0.86–0.50 percentage points for every percentage-point increase in the proportion of households >\$100k). These results suggest that utilities do not use fixed fees as a redistributive tool; rather, those with poorer customers have discounted volumetric prices across the full rate structure. These results highlight a direct equity–efficiency trade-off because utilities with more low-income households deviate more strongly from efficient price schedules by lowering marginal prices for all consumption levels.

Home age, as a proxy for infrastructure age, is correlated with marginal prices and markups, particularly for higher tiers of consumption. As the proportion of homes built after 2005 (before 1980) increases (decreases), marginal prices and markups increase (decrease). Utilities with more recently built homes might indicate higher capital costs, which are passed through.

In terms of population characteristics, utilities with greater household size (perhaps indicating more multigenerational families) have lower fixed fees and prices and larger discounts for the first consumption tier. To the extent that larger households are more likely to be low income, this result is consistent with the finding that utilities with more low-income households more severely discount marginal prices at low levels. These magnitudes appear large, but moving from the 25th to the 75th percentile would result only in a \$0.94 decrease in the fixed charge and a \$0.36/kgal decrease in price in the first price tier. An increase in the proportions of renters and non-White residents is correlated with increases in marginal prices and markups but not changes in the fixed service fee. These results contrast with the income results. Renters may not pay water bills directly, so utilities with larger proportions of renters may be able to pass through more costs. The result that utilities with greater minority populations have higher prices and markups is consistent with evidence that suggests water affordability concerns are concentrated in minority, particularly Black, communities (Cardoso and Wichman, 2022). Last, the proportion of Republican voters is positively associated with marginal prices across all tiers, but markups are not statistically different from zero. Utilities in more conservative areas have higher prices, which may capture differences in costs associated with urban versus rural areas.

Table 6: Correlates of utility rate structures and markup

	Avg. MP (\$/kgal)					Markup (% $\times$ 100)			
	Base fee	0–3 kgal	3–5 kgal	5–10 kgal	10–15 kgal	0–3 kgal	3–5 kgal	5–10 kgal	10–15 kgal
Number of customers (10,000)	-1.04*** (0.19)	0.27*** (0.07)	0.15 (0.13)	0.17 (0.14)	0.16 (0.14)	4.25*** (0.97)	2.60*** (0.65)	2.76*** (0.60)	2.74*** (0.67)
HH income: <50k (% $\times$ 100)	0.05 (0.05)	-0.03** (0.01)	-0.02 (0.01)	-0.03* (0.01)	-0.04** (0.02)	-0.66*** (0.19)	-0.77*** (0.19)	-0.96*** (0.20)	-1.21*** (0.22)
HH income: >100k (% $\times$ 100)	0.06 (0.08)	0.03 (0.02)	0.01 (0.02)	0.02 (0.02)	0.02 (0.02)	0.86*** (0.27)	0.55** (0.28)	0.54* (0.30)	0.50 (0.36)
Year built: post-2005 (% $\times$ 100)	-0.10 (0.10)	0.03 (0.02)	0.05** (0.02)	0.07*** (0.02)	0.10*** (0.03)	0.50 (0.40)	0.40 (0.34)	0.69* (0.35)	0.92** (0.41)
Year built: pre-1980 (% $\times$ 100)	-0.04 (0.03)	-0.01 (0.01)	-0.01* (0.01)	-0.02** (0.01)	-0.03*** (0.01)	-0.02 (0.11)	-0.10 (0.10)	-0.21* (0.11)	-0.34*** (0.13)
Avg. household size (persons)	-2.48* (1.49)	-0.97*** (0.34)	-0.55 (0.34)	-0.62* (0.36)	-0.53 (0.42)	-11.55** (4.75)	-2.99 (4.69)	-3.93 (4.95)	-5.43 (5.98)
Prop. renters (% $\times$ 100)	-0.01 (0.03)	0.04*** (0.01)	0.04*** (0.01)	0.05*** (0.01)	0.05*** (0.01)	0.78*** (0.14)	0.91*** (0.14)	1.05*** (0.14)	1.15*** (0.16)
Prop. non-White (% $\times$ 100)	0.01 (0.03)	0.03*** (0.01)	0.03*** (0.01)	0.02** (0.01)	0.03** (0.01)	0.44*** (0.11)	0.28*** (0.10)	0.25** (0.11)	0.26** (0.13)
Republican vote share in 2012 (% $\times$ 100)	0.02 (0.05)	0.02* (0.01)	0.03** (0.01)	0.03** (0.01)	0.03** (0.02)	0.20 (0.18)	0.27 (0.17)	0.28 (0.18)	0.31 (0.20)
Drought weeks	0.03 (0.02)	0.02*** (0.01)	0.03*** (0.01)	0.04*** (0.01)	0.04*** (0.01)	0.01 (0.10)	-0.06 (0.10)	-0.05 (0.10)	-0.03 (0.11)
1[Surface water]	0.34 (1.31)	0.43 (0.40)	1.37*** (0.37)	1.24*** (0.40)	1.29*** (0.43)	-3.49 (5.24)	1.62 (4.54)	0.04 (4.60)	1.73 (5.07)
1[Purchased water]	1.01 (1.21)	1.87*** (0.41)	2.98*** (0.38)	2.98*** (0.41)	2.98*** (0.45)	-1.13 (4.51)	-0.68 (4.19)	-2.21 (4.18)	-2.60 (4.55)
Constant	30.52*** (6.56)	6.42*** (1.74)	6.04*** (1.50)	6.47*** (1.66)	7.37*** (1.93)	4.30 (23.63)	23.35 (22.40)	38.95 (23.75)	61.15** (29.00)
Observations	6346	6346	6346	6346	6346	6346	6346	6346	6346
No. Utilities	732	732	732	732	732	732	732	732	732
R-squared	0.13	0.32	0.37	0.36	0.34	0.19	0.14	0.17	0.18
State FE:	Y	Y	Y	Y	Y	Y	Y	Y	Y
HUC6 $\times$ year FE:	Y	Y	Y	Y	Y	Y	Y	Y	Y

Notes: “HH income” is the percentage ($\times$ 100) of households with annual income below \$50k (or above \$100k). “Year built” is the percentage of homes ($\times$ 100) built after 2005 (or before 1980). “Average household size” is the average number of people per household. “Prop. renters” and “Prop. non-White” are the proportion ($\times$ 100) of residents who rent or are non-White. “Republican vote share in 2012” is the proportion ($\times$ 100) of residents in the county that voted for the Republican presidential candidate in the 2012 general election. “Drought weeks” is the annual average number of weeks in the previous 10 years that the utility experienced extreme drought or worse. Indicators for water source are also included, which means that the constant term reflects average rates/markup for utilities that rely on groundwater. Regressions are estimated on the full sample of utilities. Standard errors are clustered at the utility level.

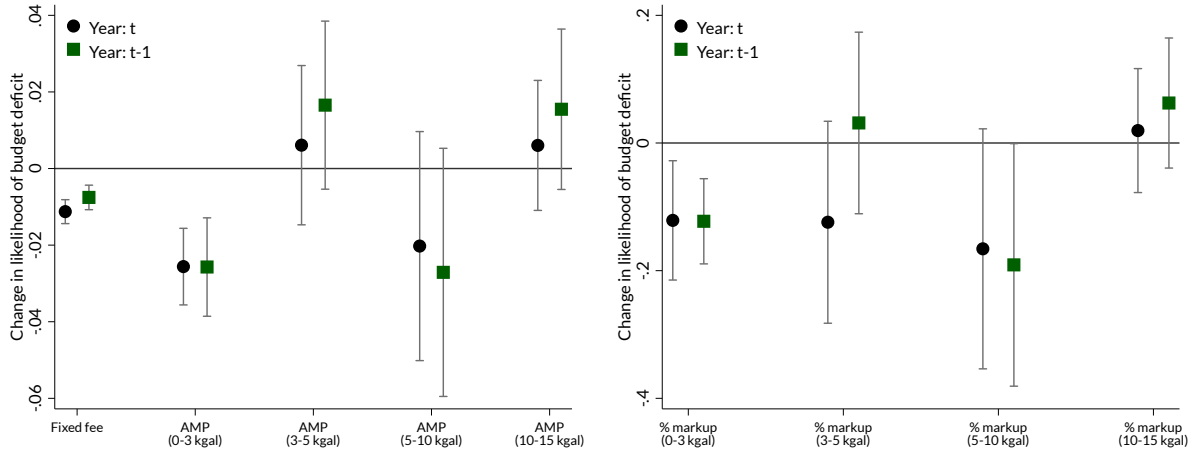


Figure 8: Changes in likelihood of budget deficit by rate structure characteristics and markup

Notes: Figures show coefficients from a linear probability model of budget deficit (= 1 if operating expenditures exceed revenues in year t) on rate structures and markup. Specifications with contemporaneous and lagged prices/markups are estimated separately. All regressions include utility and $HUC6 \times year$ fixed effects. Whiskers represent 90 percent confidence intervals.

Experience with drought is correlated with utility prices and increased rates but not markups at all levels. For every additional week of extreme drought per year, marginal prices increase by \$0.02–0.04. The utility in the 75th percentile of past drought intensity has marginal prices that are \$1.41, \$1.96, \$2.19, and \$2.52 per kgal higher than a utility in the 25th percentile. This relationship provides some suggestive evidence that water prices do reflect scarcity, although the fact that markups do not change suggests that these utilities face larger marginal costs (e.g., greater pumping costs), so water scarcity is indirectly priced into rates.

4.5 Rates, markups, and budget deficits

I perform an analysis to understand whether rate structure characteristics influence the likelihood that a utility recovers its costs. I regress a dummy variable reflecting whether operating costs exceed operating revenues on the fixed service fee, marginal prices, and percentage markup in each price tier. I include utility and $HUC6 \times year$ fixed effects, so these estimates rely on variation rates within a utility, accounting for time-varying regional shocks. I estimate the rate structure and markup regressions separately for both contemporaneous and lagged prices/markups. I present coefficients in Figure 8. In the left panel, increases in the fixed fee and average marginal prices reduce the likelihood of budget deficits, but changes in marginal prices in the higher tiers do not have a significant effect. For markups, the story is similar: the likelihood of deficits decreases as markups in the first price tier increase, with no effects from markups in higher tiers. This analysis implies that low fixed service fees and low, discounted volumetric prices in the first block are correlated with budget deficits. As discussed in Section 5, it is not necessarily welfare reducing if low fixed service fees lead to deficits, requiring cross-subsidization from alternative sources of municipal revenue. However, deficits caused by discounted volumetric prices do result in allocative efficiency costs.

4.6 Estimating revenue elasticities

Next, I focus on how changes in water rates affect annual utility revenue. Estimating aggregate revenue elasticities is useful because it reveals points along a fixed aggregate demand curve. These elasticities are policy relevant both directly, because they show how municipal revenue changes in response to price changes, and indirectly, because they reveal the slope of the demand curve.²⁸ These implied aggregate price elasticities are important inputs to understanding the welfare consequences of current pricing practices.

I adopt a panel approach to estimate revenue elasticities directly using the following equation:

$$\widetilde{\text{Revenue per customer}}_{iht} = \beta_1 \widetilde{\text{Fixed fee}}_{iht} + \beta_2 \widetilde{P}_{iht} + \alpha_i + \eta_{ht} + W'_{iht} \gamma + \varepsilon_{iht}. \quad (10)$$

The dependent variable is annual utility operating revenue divided by the number of customers. The tilde ($\widetilde{}$) represents the inverse hyperbolic sine transformation, which approximates a natural logarithm and facilitates interpretation of elasticities directly.²⁹ P_{iht} is the price of water, defined either as the average marginal (AMP_{iht}) or average (AP_{iht}) price. AMP_{iht} is the average marginal price that the typical customer faces each year (e.g., if typical consumption per household is 6 kgal per month, then AMP_{iht} would reflect the 5–10 kgal tier).³⁰ AP_{iht} is the average volumetric price for the level of consumption that the typical customer faces each year. Although, in theory, consumption should respond to marginal price, changes in revenue should also respond to changes in inframarginal prices. For similar reasons, I also estimate Equation 10 with fixed service fees included. I am primarily interested in how changes in marginal prices affect revenue and, by extension, quantity demanded. However, when utilities adjust prices or rate structures, they often also change their fixed service fee, which could confound elasticity estimates.

The coefficient of interest is $\hat{\beta}_2$, which is the percentage change in revenue as a result of a percentage change in prices. I include HUC6×year fixed effects (η_{ht}), which control for unob-

²⁸Note the correspondence between revenue elasticities and price elasticities of demand, that is, $\epsilon^{\text{Price}} = \epsilon^{\text{Revenue}} - 1$. To see this, let $R(p) \equiv pW(p)$ be total revenue from aggregate water demand $W(p)$. Define the revenue elasticity with respect to price as $\epsilon^{\text{Revenue}} = \frac{\partial R(p)}{\partial p} \frac{p}{R(p)} = \left(W(p) + p \frac{\partial W(p)}{\partial p} \right) \frac{p}{pW(p)} = 1 + \frac{\partial W(p)}{\partial p} \frac{p}{W(p)}$, where the final term is simply the price elasticity of demand, ϵ^{Price} . Spelling out this derivation makes clear that what the revenue elasticity recovers is consumer behavior in response to changes in price. This correspondence relies upon a constant elasticity of demand assumption and linear prices. In my context, water prices are nonlinear, which is an important distinction because revenue responds to both marginal and inframarginal price changes. So, approximating price elasticities of demand from revenue elasticities implies that consumers also respond to the full rate structure and not just marginal prices.

²⁹I use the inverse hyperbolic sine rather than a natural logarithm because marginal prices, specifically in the first block of consumption, can be zero. Revenue per customer is strictly positive. Using an asinh–asinh specification makes $\hat{\beta}$ a direct estimate of a revenue elasticity (Bellemare and Wichman, 2020). However, recent work has shown that the inverse hyperbolic sine is sensitive to scale (Chen and Roth, 2024; Mullahy and Norton, 2024), so I estimate the same set of revenue elasticities with natural logarithm transformations dropping any utility observations with marginal prices equal to zero. The papers focused on issues with scale-dependence and the inverse hyperbolic sine do not propose solutions for when *independent* variables contain zeros.

³⁰I also estimate a specification that includes full set of marginal prices in each utility to probe how revenue responds to the full rate schedule.

served annual shocks within each hydrological region, such as exceptionally dry years or regional economic shocks, and utility fixed effects (α_i), which control for time-invariant aspects of the municipality. I also include a vector of weather controls (W_{iht}), including number of weeks in severe drought or worse, the number of days each year maximum daily temperature falls within 10-degree C bins, and total annual precipitation. Most empirical estimates of price elasticities in water demand come from household-level data within one or several utilities or address aggregate price elasticities at the utility level cross-sectionally. My approach differs in that I have a decade of water and wastewater revenue for hundreds of water utilities, so I can estimate within-utility changes in revenue as a result of changes in water prices and rate structures, controlling for fixed, unobserved factors at the utility level and time-varying regional shocks.

As Equation 10 effectively captures changes in quantity demand, coefficients recovered from OLS estimation are likely to be biased and inconsistent because supply and demand are simultaneously determined. For municipal water pricing, this concern is less severe because prices are set administratively and fixed in the short run (Olmstead, 2009). However, two endogeneity concerns may yet arise. First, municipalities may adopt higher prices and/or a particular type of rate structure as a result of high consumption or revenue deficits, yielding potential for correlation between outcomes and errors. Second, with IBRs—or nonlinear volumetric rate structures, in general—prices directly respond to increases in quantity demanded. To deal with these concerns, I adopt an instrumental variables (IV) strategy to estimate elasticities in Equation 10.

I use two different IVs that rely on prices and rate structures for other utilities in the same HUC6 region. The first IV is generated by simulating what price the same level of consumption would be charged using the average rate structure of all other utilities in the same HUC6 region. This marginal price then serves as an instrument. This approach is similar to Wietelman et al. (2024), who use a machine-learning algorithm to predict a simulated price, but I generate a simulated price from neighboring utilities. The second IV includes the full set of averaged marginal prices in each tier for all other utilities in the same HUC6 region. This approach is similar to Olmstead (2009) and Wichman et al. (2016), although those papers use the full rate structure for utility i to instrument for its observed price. That approach may violate the exclusion restriction if consumers respond to the rate structure itself, which is particularly relevant in this setting, where inframarginal price changes affect revenue directly. I lag both sets of IVs by one year. In specifications that include fixed service fees, I include as an additional instrument the mean fixed fee across all other utilities in the same HUC6 region. The rationale for this set of IVs is that rate-setting processes might rely on information spillovers within similar geographic areas (e.g., observing a neighboring utility adopting a new rate structure) or common supply shocks. The exclusion restriction is likely satisfied because no direct way exists by which revenue in utility i in year t would be affected by rate structures in $-i$ in year $t - 1$.

In Table 7, I present the estimates for revenue elasticities. In the first column, the baseline OLS elasticity estimate is 0.14, which means that as average marginal prices increase by 10 percent, utility revenues increase by 1.4 percent. This result, although biased, is generally sensible. Water

Table 7: Revenue elasticity estimates

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Fixed fee _{iht}	0.13*** (0.03)	0.11*** (0.03)	0.15*** (0.03)	0.25*** (0.06)	0.29*** (0.08)	0.32*** (0.06)	0.31*** (0.07)
$\widetilde{AMP}_{iht}$	0.14*** (0.02)			0.52*** (0.14)	0.44*** (0.07)		
$\widetilde{AP}_{iht}$			0.31*** (0.02)			0.77*** (0.21)	0.31*** (0.05)
$\widetilde{AMP}_{iht} : 0 - 3kgal$		0.02 (0.01)					
$\widetilde{AMP}_{iht} : 3 - 5kgal$		0.09* (0.05)					
$\widetilde{AMP}_{iht} : 5 - 10kgal$		0.20* (0.11)					
$\widetilde{AMP}_{iht} : 10 - 15kgal$		0.03 (0.12)					
Observations	5085	5085	5085	5085	5085	5085	5085
No. Utilities	685	685	685	685	685	685	685
ϵ_p	-0.86	-0.80	-0.69	-0.48	-0.56	-0.23	-0.69
SE(ϵ_p)	0.02	0.11	0.02	0.14	0.07	0.21	0.05
IV: HUC6 $\widetilde{AMP}$, ($t - 1$)	—	—	—	Y	—	Y	—
IV: HUC6 mean rates, $t - 1$	—	—	—	—	Y	—	Y
Cragg-Donald Wald F stat	—	—	—	31.1	31.5	27.3	103.1

Notes: The dependent variable is the inverse hyperbolic sine of annual revenue per customer. All regressions include utility and HUC6 $\times$ year fixed effects. Standard errors are clustered at the utility level. $\widetilde{AMP}_{iht}$ is the inverse hyperbolic sine of the average marginal price for each utility-year. In Column (2), I include each average marginal price in each consumption tier. $\widetilde{AP}_{iht}$ is the average volumetric price for each utility-year. In Columns (4) and (6), $\widetilde{AMP}_{iht}$ or $\widetilde{AP}_{iht}$ is instrumented with the lagged, simulated marginal price of the mean rate structure from all other utilities in the same HUC6 region. In Columns (5) and (7), $\widetilde{AMP}_{iht}$ or $\widetilde{AP}_{iht}$ is instrumented with the mean marginal prices in each price tier for all utilities in the same HUC6 region. In Columns (4)–(7), the base service fee is instrumented with the mean base fee of all other utilities in the same HUC6 region. The inverse hyperbolic sine is applied to all base fees and price variables.

demand is inelastic, with few substitutes, which suggests that price increases should lead to revenue increases. In the second column, I include all marginal prices in each tier. Only the coefficients on the 3–5 kgal and the 5–10 kgal tiers are significant and comparable in magnitude, when averaged, to the estimate in Column 1. Because these tiers are most likely to be marginal (average household consumption is around 6 kgal per month), it make sense that these price variables have the most predictive power. Marginal prices in the higher tiers are highly collinear with lower tiers (or perfectly collinear for uniform rates). In the third column, the coefficient on average price is slightly larger and implies a revenue elasticity of 0.31. In each of these specifications, the coefficient on the base service fee is 0.11–0.15.

Turning to the IV estimates, the coefficients on average marginal price in Columns (3) and (4) are larger than their OLS counterparts, 0.52 and 0.44 for both IVs, and statistically significant. The first-stage F-statistics are around 31, suggesting that the instruments are reasonably strong. The correspondence between elasticity estimates that arise from different variation in rate-setting practices of neighboring utilities provides stronger confidence in the reliability of these estimates. For average prices in Columns (6) and (7), the estimates are 0.77 and 0.31, or a slightly wider

range, but roughly comparable in magnitude. The F-statistics are 27 and 103, indicating that the instruments, again, are reasonably strong. It is reassuring that the results are robust to different specifications of the price variable. Across all IV specifications, the coefficient on the fixed service fee is 0.25–0.32. This result is sensible: fixed fees are approximately 28 percent of the total bill at 10 kgal/month (\$25/\$89) for the average utility (see Table 1), so a 1 percent increase in the fixed fee should translate to a roughly 0.28 percent increase in total revenue, all else equal.³¹

The estimated revenue elasticity estimates imply *price* elasticities ranging from -0.23 to -0.69 , with the preferred estimates—coming from Columns (5) and (7) due to larger first-stage F-statistics—falling on the more elastic end. These estimates are within the central range of previous studies. Recent work on the price elasticity of water demand relies increasingly on high-frequency household-level data to estimate price elasticities for a single utility or a small set of utilities (e.g., Olmstead et al., 2007; Nataraj and Hanemann, 2011; Mansur and Olmstead, 2012; Wichman, 2014; Szabo, 2015; Wichman et al., 2016; El-Khattabi et al., 2021; Smith, 2022). These microdata have been useful for controlling for unobservable customer heterogeneity, revealing heterogeneity in responsiveness to price conditional on household characteristics, and understanding how consumers react to different types of rate structures. However, that research is less informative for understanding how consumers would respond in settings with different utility or demographic characteristics or what the ultimate implications of price changes would be for utility-wide outcomes. My estimates, which are generated from aggregate, annual data that control for fixed and unobservable *utility* characteristics, show that price changes affect overall utility revenues in a way that is largely consonant with estimates based on microdata. These aggregate revenue elasticities are potentially more valuable to utility managers, who often want to know how rate changes will affect their ability to recover costs.

Because the inverse hyperbolic sine is sensitive to the scale of the transformed variable (Chen and Roth, 2024; Mullahy and Norton, 2024), I re-estimate Table 7 with natural log transformations, dropping any observations with marginal prices equal to zero, and report coefficients in Table D.4 in the appendix. The results are qualitatively and quantitatively similar to those using the inverse hyperbolic sine. In particular, the revenue elasticities for $\ln(\text{AMP})$ corresponding to Columns (4) and (5) are 0.90 (SE = 0.39) and 0.44 (SE = 0.08), and the revenue elasticities for $\ln(\text{AP})$ are 0.89 (SE = 0.38) and 0.33 (SE = 0.05). Additionally, as a robustness check, I estimate price elasticities directly using the inverse hyperbolic sine of quantity consumed per account as the dependent variable in Table D.5, holding all else identical to Table 7. The corresponding IV price elasticity estimates using *AMP*, in Columns (4) and (5) of Panel B, are 1.75 (SE = 0.47) and -0.29 (SE = 0.11). For IV specifications with *AP* in Columns (6) and (7) of Panel B, the price elasticity estimates are 2.65 (SE = 0.81) and -0.18 (SE = 0.09). These results suggest that the simulated price IV in Columns (4) and (6) does poorly in removing price endogeneity with implied quantities as the dependent variable, but the IV specification in Columns (5) and (7) recovers

³¹I repeat this analysis without the base service fee in Table D.3, which produces similar results. However, including the fee is sensible when considering changes in revenue.

sensible estimates, despite the fact that quantities are not observed in the data.

5 Welfare effects of switching to marginal-cost pricing

The final piece of this analysis combines the insights from price discounts/markups with the estimated revenue elasticities to understand the welfare consequences of inefficient water pricing. I calculate the percentage change in water use for a representative customer in each utility due to switching to marginal-cost pricing, which permits calculation of the deadweight loss prevailing rates and consumer surplus and revenue changes that would result from that pricing. This analysis quantifies the magnitudes of the relevant surplus and revenue changes identified in Figure 2. For all calculations, I use estimated marginal costs and a common range of price elasticities (-0.3 to -0.7) that is consistent with the estimated revenue elasticities from Table 7.

Changes in water use when priced at marginal cost follow from utility discounts/markups and price elasticities. To simplify the analysis, I assume consumers respond to marginal price changes but do not model the effect of inframarginal price changes on demand.³² I assume the representative customer consumes the utility-specific mean each month and calculate the percentage change in consumption for each utility under a constant elasticity of demand assumption across a range of elasticities. The results are presented in Panel A of Table 8. For the median utility with IBRs, quantity demanded *increases* by 44 percent when priced at marginal cost with an elasticity of -0.5 . This occurs because IBRs mark up prices for higher consumption levels, so the average consumer experiences a decrease in marginal price. If consumers respond to average price, however, consumption decreases for the average IBR utility by 21 percent due to the severe inframarginal discounts (see Table D.6). The median utility with uniform rates sees a modest increase in quantity demanded, as prices are already close to marginal cost. For the median utility with “uniform + free” rates and utilities with “other” rates, consumption decreases by 16–25 percent due to elimination of price discounts, which lead to more dramatic consumption changes with average price responsiveness, given the large change in inframarginal rates. These results highlight the allocative inefficiencies inherent in nonlinear rate structures and, as exhibited by utilities with uniform rates, the relative efficiency of marginal-cost pricing.

The allocative costs of inefficient pricing are quantified in Panel B of Table 8. Inefficient pricing generates welfare losses because observed and optimal prices diverge, leading to subop-

³²Inframarginal price changes could affect demand via virtual income effects that arise from kinked budget constraints. Because the unit of analysis is a representative consumer for each utility paired with a rate structure, I do not estimate demand or evaluate welfare within a structural discrete-continuous choice (DCC) framework that accounts for the joint decision of where on the price schedule to locate and, conditional on that choice, how much to consume (e.g., Hanemann, 1984; Olmstead et al., 2007; Szabo, 2015; Wang and Wolak, 2022). Recent research questions the whether consumers actually understand nonlinear rate structures, which casts doubt on the validity of the DCC approach in these settings (e.g., Ito, 2014; Wichman, 2014; Brent and Ward, 2019; Shaffer, 2020; Ito and Zhang, 2023). To account for consumers potentially responding to inframarginal prices, I replicate the welfare analysis assuming that they respond to average volumetric rather than marginal prices: the median utility with a nonlinear rate structure sees substantial declines in consumption due to the severely discounted inframarginal prices. I present marginal price responsiveness here because it is inherently conservative. Both frameworks produce similar estimates of surplus loss and revenue gains because those operate through inframarginal channels.

Table 8: Welfare effects of switching to marginal-cost pricing—marginal price responsiveness

Panel A: Percentage change in quantity demanded ($\% \times 100$)					
	$\epsilon_p = -0.3$	$\epsilon_p = -0.4$	$\epsilon_p = -0.5$	$\epsilon_p = -0.6$	$\epsilon_p = -0.7$
IBR	25.4	33.9	42.4	50.9	59.4
Uniform	6.4	8.6	10.7	12.9	15.0
Uniform + free	-10.2	-13.6	-17.0	-20.4	-23.8
Other	-14.5	-19.4	-24.2	-29.0	-33.9
Total	8.4	11.2	14.0	16.8	19.6

Panel B: Annual deadweight loss from inefficient pricing (\$/customer)					
	$\epsilon_p = -0.3$	$\epsilon_p = -0.4$	$\epsilon_p = -0.5$	$\epsilon_p = -0.6$	$\epsilon_p = -0.7$
IBR	13.3	17.8	22.2	26.7	31.1
Uniform	4.2	5.5	6.9	8.3	9.7
Uniform + free	8.2	11.0	13.7	16.5	19.2
Other	16.2	21.6	27.0	32.4	37.8
Total	9.2	12.3	15.3	18.4	21.5

Panel C: Annual change in consumer surplus (\$/customer)					
	$\epsilon_p = -0.3$	$\epsilon_p = -0.4$	$\epsilon_p = -0.5$	$\epsilon_p = -0.6$	$\epsilon_p = -0.7$
IBR	-26.0	-19.8	-18.9	-14.7	-10.8
Uniform	13.0	13.0	13.0	13.1	13.1
Uniform + free	-158.7	-153.3	-146.7	-141.9	-134.2
Other	-32.6	-28.0	-23.3	-18.6	-14.0
Total	-54.0	-52.5	-51.4	-50.3	-47.5

Panel D: Annual change in municipal revenue (\$/customer)					
	$\epsilon_p = -0.3$	$\epsilon_p = -0.4$	$\epsilon_p = -0.5$	$\epsilon_p = -0.6$	$\epsilon_p = -0.7$
IBR	40.4	41.8	37.6	30.4	17.6
Uniform	-9.0	-7.8	-5.4	-2.1	-5.7
Uniform + free	136.2	116.7	93.7	85.8	85.1
Other	34.5	17.2	29.1	12.4	-4.2
Total	55.5	54.6	42.4	32.7	22.9

Notes: Entries are annual statistics in 2015 for the representative customer in the median utility. The results include both combined water and wastewater utilities and water-only utilities. The representative customer is assumed to consume utility-specific mean quantity supplied each month of the year. For context, annual billed amounts under marginal-cost pricing (with $\epsilon = -0.5$) for the average utility are \$953, \$877, \$818, and \$775 for utilities that use IBRs, uniform, uniform + free, and other rates.

timal consumption. I calculate deadweight loss (DWL) with the change in quantity demanded (Δw) that corresponds with each elasticity estimate and use the Harberger approximation, where $DWL \approx 0.5|\Delta w(p - MC)|$. The annual DWL cost of inefficient pricing is approximately \$7–\$27 per customer across all rate structures with an elasticity of -0.5 . Uniform rates have the smallest allocative costs; nonlinear rates have larger costs. DWL increases, sensibly, for more elastic demand. DWL quantities are similar under average cost pricing (see Table D.6).

As demonstrated in Figure 2, switching from nonlinear rates to marginal-cost pricing generates ambiguous welfare effects, and correcting nonlinear rates results in a transfer of consumer surplus to municipal revenue. In Panel C of Table 8, I calculate the change in consumer surplus as the area between the old, potentially nonlinear, rate structure $p(w)$ and the estimated marginal cost up to initial consumption (w_0) or efficient consumption (w^*), whichever is greater, less the amount of surplus change that arises from demand responses: $\Delta CS = \int_0^{\max[w_0, w^*]} [p(w) - MC] dw - X$. The integral accounts for the difference between prices and

marginal costs across the distribution of consumption, and the area X accounts for surplus changes that arise through demand responses, which can be approximated by the area of DWL (see Figure 2). I calculate ΔCS using the average discounts/markup in each price tier and present these quantities in Panel C of Table 8. Utilities with IBRs have modest reductions in surplus, totaling around \$14 per customer per year. With IBRs, the inframarginal loss in CS is tempered by increases in CS due to increases in quantity demanded (as illustrated in Panel B of Figure 2). Utilities with uniform rates see increases in consumer surplus of \$14 per customer per year. However, utilities with “uniform + free” rates exhibit substantial decreases in consumer surplus, averaging around \$146 per customer annually at a 0.5 elasticity, which makes sense: these consumers are faced with the elimination of a large discount on inframarginal consumption. Because >75 percent of utilities have either IBRs or “uniform + free” rates, consumer surplus losses are prevalent across the sample.

Finally, I calculate municipal revenue changes as a result of marginal-cost pricing. Even with inelastic demand and price increases implied by marginal-cost pricing, the nonlinearity of water rates leads to *ex ante* ambiguous changes in revenue. I first calculate volumetric revenue that arises from the initial price structure and initial levels of consumption. I then calculate volumetric revenue under marginal-cost pricing, accounting for the demand response. The difference between the two quantities is the change in revenue reported in Panel D of Table 8. For all utilities except those with uniform rates, revenue increases. Utilities with IBRs or other rates generate about \$29–\$37 more revenue per customer annually at a 0.5 elasticity, which decreases as demand becomes more elastic. Due to close-to-marginal-cost pricing for utilities with uniform rates, they see a small decrease in revenue. Utilities with “uniform + free” rate structures see substantial revenue gains, averaging around \$91 per customer annually at a 0.5 elasticity. The average utility has approximately 5,000 customers, which means that the typical utility with “uniform + free” could increase revenue by nearly a half-million dollars per year by removing the free allocation of water. Of course, this change would raise inframarginal prices and generate distributional effects, but, as discussed, equity objectives should not be targeted through volumetric subsidies. In practice, utilities could reduce the fixed service fee by a modest amount (\$5–10 per month) to keep rates revenue neutral, make the fixed service fee progressive (e.g., Borenstein et al., 2021; Burger et al., 2020; Wietelman et al., 2024), or lower the fixed service fee and cover fixed costs with alternative sources of municipal revenue.

For the median utility across all characteristics, however, marginal-cost pricing corrects deadweight loss from inefficient pricing and raises revenue, regardless of how consumers respond to nonlinear prices (e.g., average vs. marginal price). At a central elasticity, revenue gains come close to offsetting consumer surplus losses. For more elastic demand, revenue gains shrink and consumer welfare losses increase, along with a concomitant increase in deadweight loss. This suggests that if utilities are motivated, in part, by maximizing consumer surplus, nonlinear pricing does so at a comparatively high cost from foregone revenue and deadweight loss. These averages mask dramatic heterogeneity across pricing practices and utility characteristics: varia-

tion within types of rate structures could lead to differing implications for social welfare.

6 Discussion and conclusion

Overall, this paper characterizes the efficiency costs of municipal water pricing practices. First, I develop a modeling framework as a benchmark for understanding how municipal utilities should design two-part prices when balancing efficiency, equity, and cost-recovery. Empirically, I use a large-scale data set of more than 700 water utilities across the Southeast to provide descriptive evidence of how water prices deviate from the theoretical ideal. I show that many utilities fail to balance their budget exactly with revenue raised from water rates, with some running deficits and others surpluses. I then estimate marginal costs of water and wastewater service, which shows that water rates are generally marked down from their efficient level for low levels of consumption, although markups have considerable heterogeneity. I provide evidence that these discounts are larger for utilities with larger low-income populations and lead to budget shortfalls, suggesting a direct equity–efficiency trade-off. Water scarcity works in the opposite direction, leading to higher prices on average. Finally, I estimate revenue elasticities in an IV framework that controls for unobserved factors at the utility level using aggregate data. For every percentage point increase in price, utility revenues increase by 0.5–0.7 percentage points. I conclude by illustrating the utility-wide welfare consequences of imposing marginal-cost pricing for all units of water. The average utility could increase revenue and eliminate allocative inefficiencies by adopting uniform marginal prices that reflect marginal costs. To address concerns about equity, any revenue that is needed to cover fixed costs could be recovered through fixed fees proportional to income, or municipalities could draw from less regressive sources of municipal revenue.

Several shortcomings of this analysis are worth noting. One issue concerns the objective function of municipal water utilities. Assuming that net-surplus maximization is the primary goal is a simplification. In reality, utilities may have other objectives, such as maintaining affordability, ensuring reliability, or responding to political incentives (e.g., keeping prices low to keep voters happy), which may result in different pricing rules. (Timmins, 2002b), for example, demonstrates that discounted marginal prices can be supported by an objective function that weights consumer surplus gains more heavily than revenue or net-surplus maximization, which could be driven by voters’ preferences for redistribution (Timmins, 2002a). I show that discounting volumetric prices generates social welfare losses relative to using lump-sum transfers (i.e., progressive fixed fees) or alternative revenue sources to recover fixed costs, but that result is predicated on net-surplus maximization. Future research on optimal two-part tariffs under alternative objective functions would be valuable to better understand the political economy of water pricing.

Second, practical or political obstacles may prevent utilities from adopting efficient prices. Recovering fixed costs more progressively could be done through income-graduated fixed fees or alternative tax revenue. The former would require information on (a proxy for) household income, which may be difficult to obtain and could require changing municipal codes. Drawing

from general fund revenue to recover some portion of fixed costs is more feasible, as intragovernmental transfers are common, although this could increase distortionary taxes elsewhere. Utilities do, however, implement nonlinear rates for redistributive purposes, either across the consumption distribution (i.e., disincentivizing low-value outdoor use) or income distribution (i.e., providing discounts for essential needs). These practices may be relatively ineffective and inefficient, so it is worth understanding the social costs of these constraints.

Last, my empirical focus is on the Southeastern United States, which may not be representative of supply issues in other geographies. The conceptual trade-offs are generalizable, however, and the rate structures and price levels in the Southeast are similar to those in other regions. Arid regions that rely on increasingly depleted groundwater stocks may have differential cost structures with important intertemporal trade-offs. Moreover, there may be temporal variation in costs within days and across seasons that is not captured here. Future research should explore how these dynamic efficiency considerations map to optimal rate structures.

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Online Appendix

Efficiency, Equity, and Cost-Recovery Trade-Offs in Municipal Water Pricing

Casey J. Wichman

Appendix A Detailed conceptual framework

As a starting point, my framework largely follows that of Levinson and Silva (2022)—which begins as a formal model of Coase (1946)—except refocused from a regulator’s electricity pricing decisions to water pricing where the municipal government maintains a water utility.

A.1 Household’s problem

Consider a household that maximizes utility over water consumption, w^i , and a numeraire good x^i , where superscripts index households. The household earns income M^i , which is subject to a marginal tax rate of $\tau \geq 0$ and a lump-sum government transfer g^i (both of which the household takes as given). The household spends income on water (at marginal price p with a fixed monthly access fee $t \geq 0$) and the numeraire good (with price normalized to 1) in the present period, leading to the budget constraint: $pw^i + t + x = (1 - \tau)M^i + g^i$. Let $\hat{M}^i \equiv (1 - \tau)M^i + g^i - t^i$ represent after-tax residual income. The household’s problem becomes

$$\max_{w_i} u^i(w_i, \hat{M}^i - pw^i). \quad (\text{A.1})$$

The household’s first-order condition is

$$\frac{u_w^i}{u_x^i} = p \text{ for all } i = 1, \dots, n. \quad (\text{A.2})$$

This condition implicitly defines demand functions for water and the numeraire good: $w^i(p, \hat{M}^i)$ and $x^i(p, \hat{M}^i) = \hat{M}^i - pw^i(p, \hat{M}^i)$. Plugging these demand equations into the household’s utility function results in an indirect utility function: $v^i(p, \hat{M}^i) = u(w^i(p, \hat{M}^i), x^i(p, \hat{M}^i))$ for all $i = 1, \dots, n$.

A.2 Government’s problem

The government chooses prices, access fees, and marginal tax rates to maximize the sum of indirect utilities that households receive. It manages the water utility that has fixed costs F and per-unit costs w . So, its total cost of supplying $W = \sum_i w^i$ units of water is $F + c \sum_i w^i$. The government must balance its budget each year, resulting in the following budget constraint:

$$\sum_i [t + pw^i + \tau M^i] = F + c \sum_i w^i + G, \quad (\text{A.3})$$

where $G = \sum_i g_i$ are total governmental transfers. The left-hand side of this equation is governmental revenue; the right-hand side are governmental expenditures.

A.3 A self-sufficient water utility

First, assume the government chooses p and t but treats τ as fixed and exogenous and that $\tau \sum_i M^i = G$, so that it focuses only on balancing revenue and costs for its utility. This results in the simpler budget constraint: $\sum_i [t + pw^i - cw^i] = F$. The government’s problem becomes

$$\max_{p, t} \sum_i v^i(p, \hat{M}^i) + \lambda \left[\sum_i [t + pw^i - cw^i] - F \right] \text{ for } i = 1, \dots, n, \quad (\text{A.4})$$

where $\hat{M}^i = M^i - t$. This problem results in the following first-order conditions:

$$\frac{\partial v^i}{\partial p} + \lambda[w^i + (p - c)w_p^i] = 0 \quad (\text{A.5})$$

$$-\frac{\partial v^i}{\partial \hat{M}} + \lambda[1 - (p - c)w_{\hat{M}}^i] = 0. \quad (\text{A.6})$$

Combining these conditions to remove the Lagrangian multiplier results in

$$-\frac{\partial v^i / \partial p}{\partial v^i / \partial \hat{M}} = \frac{w^i + (p - c)w_p^i}{1 - (p - c)w_{\hat{M}}^i} \text{ for all } i \quad (\text{A.7})$$

The left-hand side of this equation equals w^i (for all i) based on Roy's identity, so the equation can be rewritten as

$$(p - c)[w^i w_{\hat{M}}^i + e_p^i] = 0. \quad (\text{A.8})$$

The term in brackets is the derivative of the Hicksian demand function for water, by the Slutsky equation, which is strictly negative. That is, $h_p^i \equiv \partial h^i(p, u) / \partial p < 0$, so,

$$(p - c)h_p^i = 0 \implies p = c. \quad (\text{A.9})$$

The optimal marginal price of water will be set equal to marginal cost.

Substituting the result that $p = c$ into the government's budget constraint results in setting fixed access fees equal to average fixed costs, or, $t = F/n$.

This model follows Levinson and Silva (2022) almost exactly and reproduces the canonical result of Coase (1946) in that marginal prices should be set equal to marginal costs and fixed costs should be allocated evenly across customers.

Notably, if the government can set individualized prices, p_i —which might reflect price discrimination across customers or heterogeneous marginal prices across the consumption distribution (i.e., increasing block prices)—the insight of this model does not change. The government's first-order conditions in equations A.5 and A.6 would become

$$\frac{\partial v^i}{\partial p} + \lambda[w^i + (p_i - c)w_p^i] = 0 \text{ for all } i \quad (\text{A.10})$$

$$-\frac{\partial v^i}{\partial \hat{M}} + \lambda[1 + (p_i - c)w_{\hat{M}}^i] = 0 \text{ for all } i \quad (\text{A.11})$$

Following the same steps would result in the condition that $p_i = c$ for all i , or that all units of consumption should be charged the same marginal price.

Appendix B Detailed data description

B.1 Sample restrictions

The raw rates data have 806 water systems in Georgia and 697 utilities in North Carolina, totaling 1,503 utilities. After merging the rates data with municipal financial data with fuzzy string matching with manual inspection and correction, I have 966 utilities. This removes private utilities that would not be present in the municipal finance data and utilities that could not be matched with that data due to missing data or incorrect identifiers. To arrive at the final sample of 732 utilities, I impose the following series of sample restrictions and data quality checks:

- Utilities that only provide wastewater service are dropped (39 utilities).
- Utilities with fewer than 8 annual observations are dropped (93 utilities in Georgia and 58 in North Carolina).
- Observations are replaced with missing values if total water bills at 3, 5, 10, or 15 kgal/month are smaller than the fixed service fee. This is primarily an issue with Georgia rates data in 2019. (Obs. = 126)
- Observations are replaced with missing values if the base water service fee changes by more than +/- 80 percent year over year. This is primarily an issue with Georgia rates data in 2019. (Obs. = 58)
- Observations are replaced with missing values if total wastewater bills at 3, 5, 10, or 15 kgal/month are smaller than the fixed service fee. This is primarily an issue with Georgia rates data in 2019. (Obs. = 87)
- Observations are replaced with missing values if the base wastewater service fee changes by more than +/- 80 percent year over year. This is primarily an issue with Georgia rates data in 2019. (Obs. = 54)
- Revenue or expenditure observations reported as zero are replaced with missing values. (Obs. = 6 + 13)
- Year-to-year changes in total revenue or total expenditures that exceed +/- 100 percent are replaced with missing values. (Obs. = 43 + 78)
- Observations with average quantities of water supplied that exceed 300 kgal per year per customer are replaced with missing values. These likely stem from incorrect reporting of number of customer accounts. (Obs. = 223)
- Outliers (observations > 95th percentile) in terms of average costs are replaced with missing values. (Obs. = 359)
- Outliers (observations > 95th percentile) in terms of average revenue are replaced with missing values. (Obs. = 501)
- All observations for City of Atlanta in 2008 are removed due to data irregularities.

B.2 Calculating implied quantities supplied

In the annual financial data, I observe total operating revenues (TR). In the water rates surveys, I observe bills at different levels of consumption. Total annual revenue can be defined as follows:

$$TR = n \times \text{Avg. annual bill} = nt + p \sum_i w_i, \quad (\text{B.1})$$

where TR is annual water system revenue, n is the number of customer accounts, p is the volumetric price of water, and w_i is the quantity of water consumed by customer i . This formulation assumes a constant volumetric price, which I approximate with an average volumetric price to incorporate the nonlinearity of rate structures (e.g., increasing block rates).

By rearranging, I can express total water supplied (W) for a given utility each year:

$$W \equiv \sum_i w_i = \frac{TR - nt}{p}. \quad (\text{B.2})$$

To construct an empirical analog of Equation B.2, I use the following equation:

$$\tilde{W} = \frac{TR - 12n\text{Bill}_{0\text{kgal}}}{(\text{Bill}_{10\text{kgal}} - \text{Bill}_{0\text{kgal}})/10}, \quad (\text{B.3})$$

where $\text{Bill}_{0\text{kgal}}$ is the monthly fixed service fee, or total bill evaluated at zero consumption, and $\text{Bill}_{10\text{kgal}}$ is the total bill at 10 kgal per month. The difference in bills is divided by 10 to put the denominator in units of \$/kgal. The denominator is an estimate of the average volumetric price of water for consumption up to 10 kgal/month.

The typical customer in the Southeast consumes 5–10 kgal per month (see, e.g., Wichman et al., 2016; Ferraro and Price, 2013; Wichman and Ferraro, 2017; Wichman, 2014, 2017). Figure B.1 shows the distribution of implied water consumption per household-month for 2015 following the earlier calculation. The distribution is right-skewed, with a median of 5.7 kgal/month and a mean of 6.8 kgal/month across the full sample, and consistent with typical water consumption patterns in the Southeast, where the median household consumes 5–10 kgal per month.

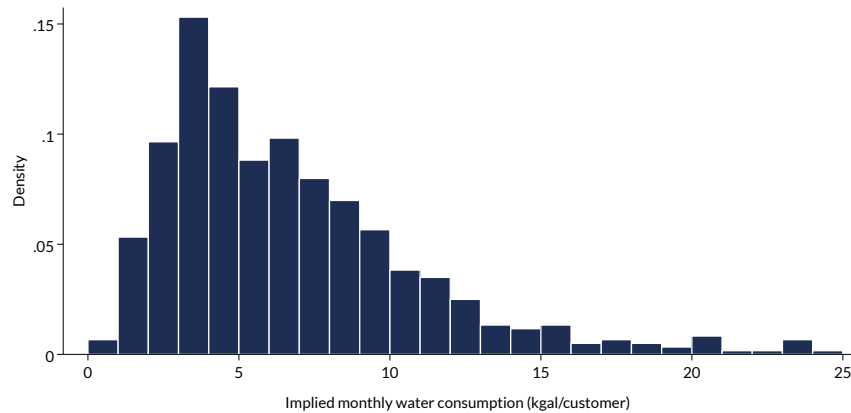


Figure B.1: Distribution of implied water consumption per household-month, 2015

A utility's total revenue contains more than residential consumption. Generally, however,

residential bills constitute the bulk of operating revenues, and the vast majority of customers are residential. Although commercial and industrial customers are also present and often charged on a different rate structure, they typically use more water per month than the average household but pay a lower average price. As a result, the implied level of water supply in Equation B.3 is a reasonable approximation of total water use, but it is subject to measurement error.

Appendix C Testing average cost pricing

To test whether utilities price water at average costs more formally, I regress average total bills at 5 kgal per month (that is, average billed amounts divided by quantity, which reflects an average price inclusive of fixed service fees) on annual average total costs (i.e., $AC_{iht} = TC_{iht}/w_{iht}$). I choose 5 kgal/month because it reflects slightly below the within-sample median level of implied monthly consumption. The regression takes the form

$$\text{Avg. Bill}_{iht}^{5\text{kgal}} = \alpha + \beta AC_{iht} + \eta_{ht} + \varepsilon_{iht}, \quad (\text{C.1})$$

where η_{ht} is a year fixed effect interacted with each six-digit hydrological unit (representing common river basins) and ε_{iht} is the residual error term. I include weights to account for the number of accounts within each utility. The estimated coefficient $\hat{\beta}$ provides an ad-hoc test for how much changes in average utility costs reflect changes in average prices. $\hat{\beta} = 1$ would provide evidence in favor of average cost pricing. In other words, increases in average utility costs lead to dollar-for-dollar increases in average bills. If $\hat{\beta} < 1$, this would suggest that utilities fail to fully pass through costs onto consumers. Last, if $\hat{\beta} > 1$, then average bills rise more than proportionally to changes in utility costs.

The results for estimation of Equation C.1 are presented in Table C.1. The coefficient estimates are small, ranging from 0.62 in Georgia to 0.41 in North Carolina and averaging 0.53 for the full sample (Column 3). They suggest that for every dollar that costs increase, utilities pass through \$0.53 on the typical water bill at 5 kgal/month. Utility size is heterogeneous. Small utilities (fewer than 1,000 customers) pass through approximately one-quarter of changes in cost to customers (26 cents on the dollar). This is 33 percent for medium utilities and 70 percent for large utilities. A coefficient of 1 would suggest that, on average, utilities price water with respect to average cost. Additionally, this relationship is even weaker when weighting each utility equally, and the results are nearly identical with lagged average cost, which captures potential delays in the rate-setting process (see Tables C.2, C.3, and C.4).

Table C.1: Average cost pricing

	GA only	NC only	Full	Small	Medium	Large
Average cost (\$/kgal)	0.62*** (0.10)	0.39*** (0.03)	0.53*** (0.08)	0.26*** (0.02)	0.33*** (0.03)	0.69*** (0.08)
Observations	3265	3106	6376	2350	2568	1430
No. Utilities	381	350	732	303	278	150
R-squared	0.72	0.54	0.66	0.57	0.52	0.78
HUC6-by-Year FE?	Y	Y	Y	Y	Y	Y

Notes: The dependent variable is the average monthly bill (evaluated at 5 kgal/month) at the utility-by-year level. Regressions include year-by-six-digit-hydrological-unit fixed effects. Standard errors are clustered by utility. Regression is weighted number of accounts in each utility. A coefficient of 1 indicates that increases in average bills at 10 kgal/month lead to dollar-for-dollar increases in average total costs.

Table C.2: Average cost pricing (unweighted)

	GA only	NC only	Full	Small	Medium	Large
Average cost (\$/kgal)	0.22*** (0.02)	0.29*** (0.02)	0.26*** (0.02)	0.23*** (0.02)	0.31*** (0.03)	0.51*** (0.06)
Observations	3265	3106	6376	2350	2568	1430
No. Utilities	381	350	732	303	278	150
R-squared	0.29	0.47	0.48	0.60	0.52	0.60
HUC6-by-Year FE?	Y	Y	Y	Y	Y	Y

Notes: The dependent variable is the average monthly bill (evaluated at 5 kgal/month) at the utility-by-year level. Regressions include year-by-six-digit-hydrological-unit fixed effects. Standard errors are clustered by utility. Regression is weighted number of accounts in each utility. A coefficient of one indicates that increases in average bills at 10 kgal/month lead to dollar-for-dollar increases in average total costs.

Table C.3: Average cost pricing (lagged, weighted)

	GA only	NC only	Full	Small	Medium	Large
Lagged average cost (\$/kgal)	0.64*** (0.10)	0.42*** (0.03)	0.56*** (0.08)	0.26*** (0.03)	0.35*** (0.03)	0.70*** (0.07)
Observations	2538	2607	5147	1746	2144	1227
No. Utilities	360	344	705	283	273	148
R-squared	0.72	0.56	0.67	0.55	0.52	0.78
HUC6-by-Year FE?	Y	Y	Y	Y	Y	Y

Notes: The dependent variable is the average monthly bill (evaluated at 5 kgal/month) at the utility-by-year level. Regressions include year-by-six-digit-hydrological-unit fixed effects. Standard errors are clustered by utility. Regression is weighted number of accounts in each utility. A coefficient of one indicates that increases in average bills at 10 kgal/month lead to dollar-for-dollar increases in average total costs.

Table C.4: Average cost pricing (lagged, unweighted)

	GA only	NC only	Full	Small	Medium	Large
Lagged average cost (\$/kgal)	0.22*** (0.03)	0.31*** (0.02)	0.27*** (0.02)	0.23*** (0.02)	0.33*** (0.03)	0.53*** (0.06)
Observations	2538	2607	5147	1746	2144	1227
No. Utilities	360	344	705	283	273	148
R-squared	0.26	0.47	0.46	0.59	0.51	0.60
HUC6-by-Year FE?	Y	Y	Y	Y	Y	Y

Notes: The dependent variable is the average monthly bill (evaluated at 5 kgal/month) at the utility-by-year level. Regressions include year-by-six-digit-hydrological-unit fixed effects. Standard errors are clustered by utility. Regression is weighted number of accounts in each utility. A coefficient of one indicates that increases in average bills at 10 kgal/month lead to dollar-for-dollar increases in average total costs.

Appendix D Additional tables and figures

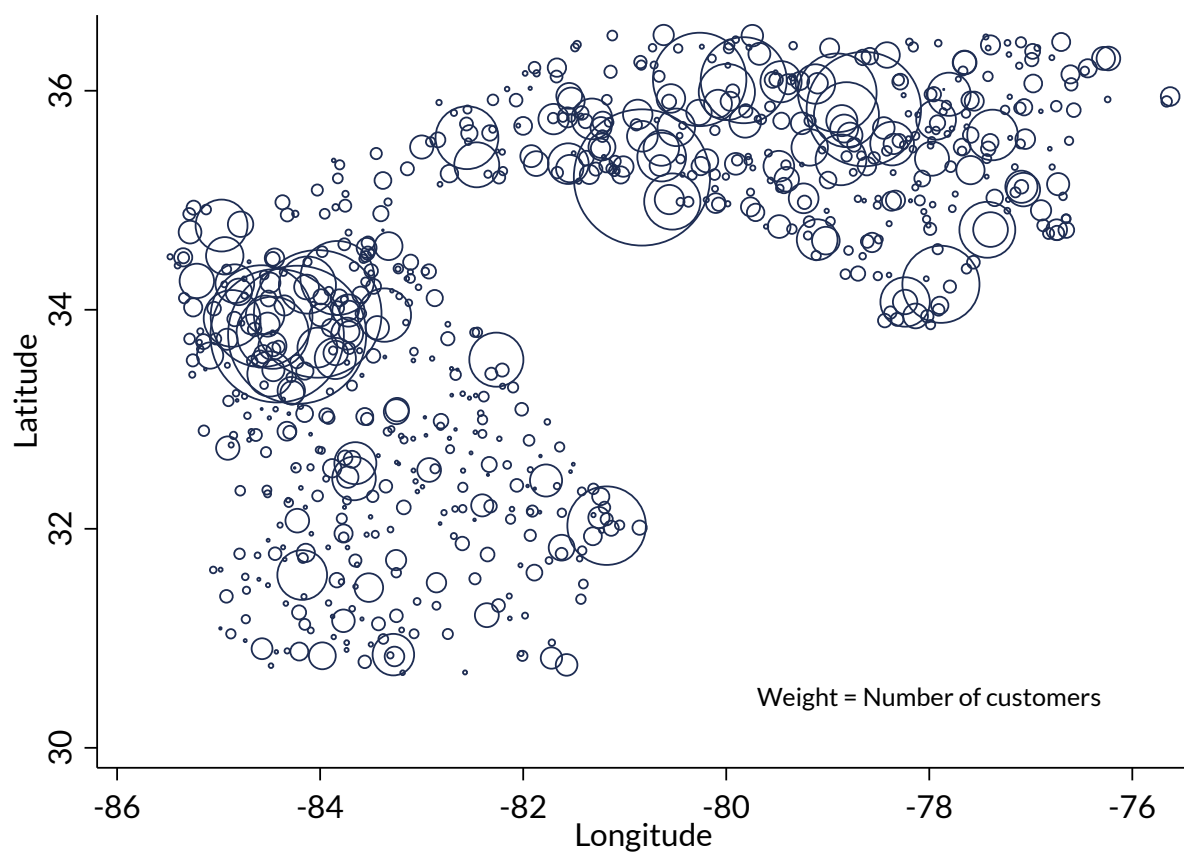


Figure D.1: Location of utilities weighted by number of customers

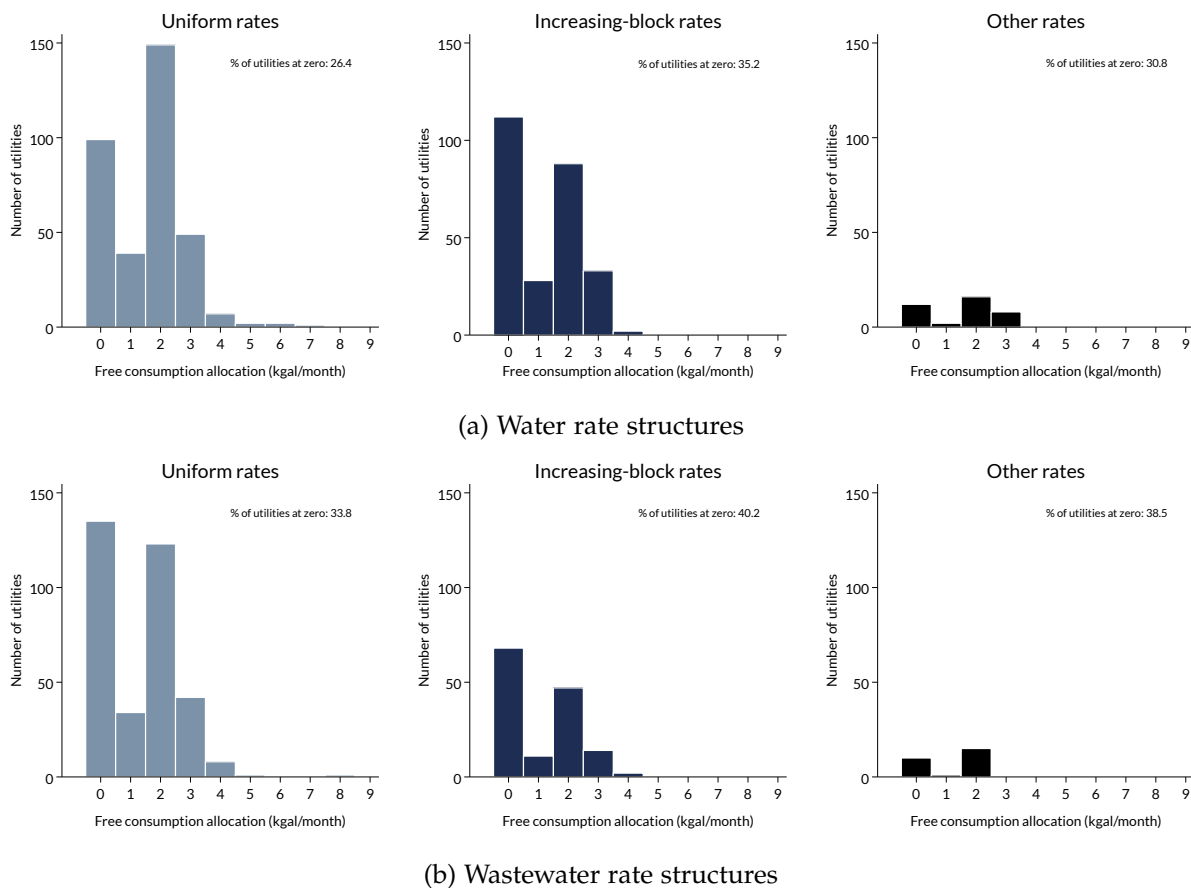


Figure D.2: Volumetric consumption included in fixed service fee at zero marginal cost

Notes: Plots represent last observed rate structure for each utility.

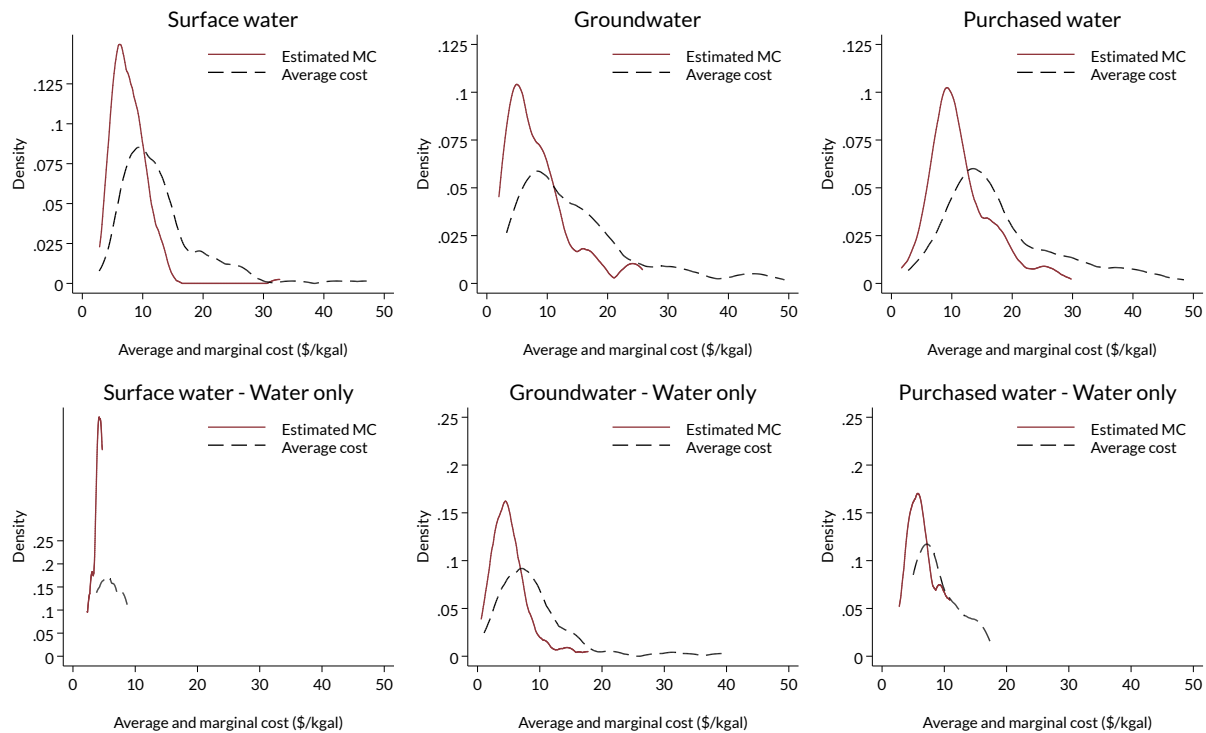


Figure D.3: Average and estimated marginal costs using trans-log-like cost function approach

Notes: Marginal costs are estimated from Columns (4)–(6) in Table 4 with HUC6-by-year-by-utility size fixed effects. Data presented are for 2015.

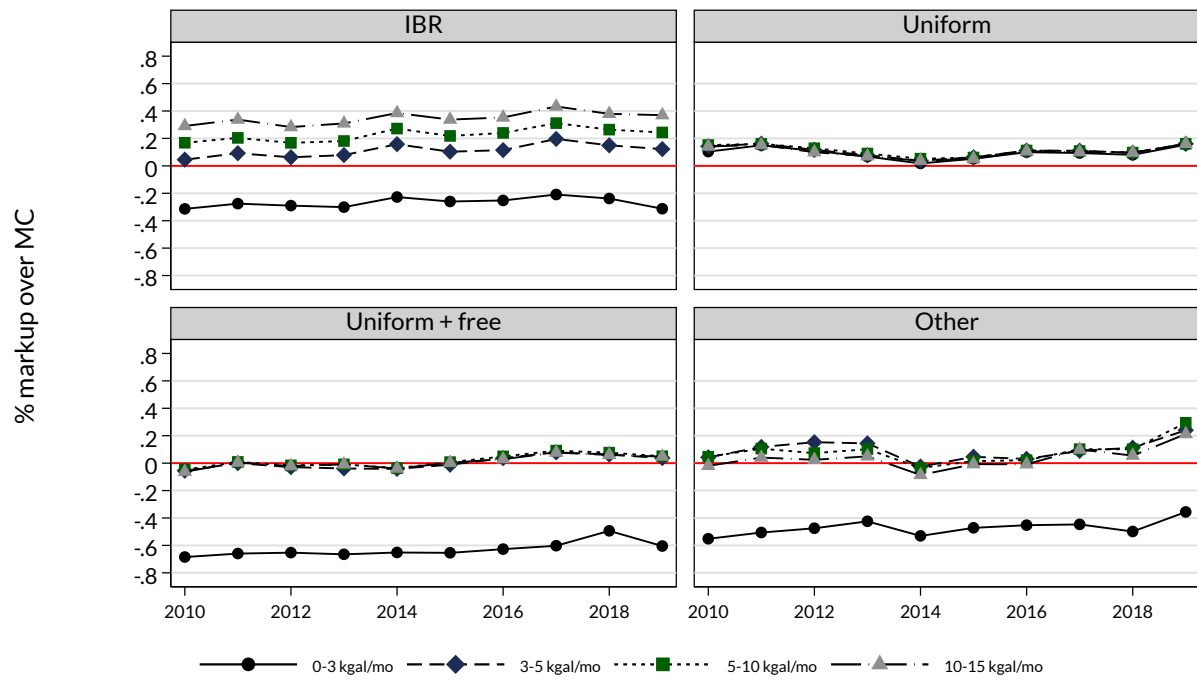


Figure D.4: Percent markup over time by rate structure

Table D.1: Summary statistics by state

Panel A: Georgia utilities						
	Mean	SD	10th %ile	50th %ile	90th %ile	N
Annual water supplied (million gallons)	0.6	2.0	0.0	0.1	0.9	382
Number of customers	5,060.1	20,403.5	166.0	1,070.5	7,825.0	382
Avg. monthly usage (kgal/month/customer)	8.5	5.8	2.1	6.9	17.3	382
Operating ratio (revenue/expenditures)	1.1	0.4	0.7	1.0	1.5	382
Combined water/wastewater utilities (%)	73.6	44.2	0.0	100.0	100.0	382
Source: Groundwater (%)	64.7	47.9	0.0	100.0	100.0	382
Source: Surface water (%)	17.5	38.1	0.0	0.0	100.0	382
Source: Purchased water (%)	17.8	38.3	0.0	0.0	100.0	382
Total bill: 5 kgal (\$)	42.3	16.8	21.7	40.4	63.1	382
Total bill: 10 kgal (\$)	71.7	32.5	35.1	65.7	114.1	382
Total bill: 15 kgal (\$)	102.0	50.8	47.0	93.3	172.2	382
Fixed service fee (\$)	22.4	9.9	10.7	21.4	35.3	382
Base free water allocation (kgal/month)	1.6	1.2	0.0	2.0	3.0	382
Base free sewer allocation (kgal/month)	1.4	1.4	0.0	2.0	3.0	281
Avg. marginal price: 0–3 kgal (\$/kgal)	2.9	2.7	0.0	2.1	6.8	382
Avg. marginal price: 3–5 kgal (\$/kgal)	5.6	3.2	2.3	4.9	10.0	382
Avg. marginal price: 5–10 kgal (\$/kgal)	5.9	3.5	2.4	5.0	10.5	382
Avg. marginal price: 10–15 kgal (\$/kgal)	6.1	3.9	2.4	5.0	11.0	382
Avg. total bill: 5 kgal (\$/kgal)	8.5	3.4	4.3	8.1	12.6	382
Avg. total cost (\$/kgal)	10.0	7.9	3.1	7.7	19.3	382
Median household income (\$ 1,000s)	36.8	13.2	22.0	34.2	55.5	382
Median home value (\$ 1,000s)	112.4	55.2	57.3	102.0	172.5	382
Pct. owner occupied	62.6	13.4	45.9	62.4	80.4	382
Prop. White (%)	61.2	21.6	32.6	61.3	91.0	382
Prop. Black (%)	33.7	21.8	4.8	32.9	63.9	382
Prop. Hispanic (%)	5.8	7.9	0.0	3.3	13.5	382
No. weeks with severe drought, 2010-2019	113.0	22.3	80.0	117.0	137.0	382

Panel B: North Carolina utilities						
	Mean	SD	10th %ile	50th %ile	90th %ile	N
Annual water supplied (million gallons)	0.5	1.9	0.0	0.1	1.0	350
Number of customers	6,558.4	19,955.8	325.0	1,622.0	11,914.0	350
Avg. monthly usage (kgal/month/customer)	5.8	4.0	2.0	4.8	11.1	350
Operating ratio (revenue/expenditures)	1.0	0.2	0.8	1.0	1.3	350
Combined water/wastewater utilities (%)	86.3	34.4	0.0	100.0	100.0	350
Source: Groundwater (%)	38.9	48.8	0.0	0.0	100.0	350
Source: Surface water (%)	29.7	45.8	0.0	0.0	100.0	350
Source: Purchased water (%)	31.4	46.5	0.0	0.0	100.0	350
Total bill: 5 kgal (\$)	61.7	21.8	36.6	58.6	91.5	350
Total bill: 10 kgal (\$)	107.4	39.5	61.0	104.3	160.0	350
Total bill: 15 kgal (\$)	154.1	59.9	82.9	148.6	231.7	350
Fixed service fee (\$)	28.2	14.2	11.6	27.1	46.7	350
Base free water allocation (kgal/month)	1.3	1.2	0.0	1.5	3.0	350
Base free sewer allocation (kgal/month)	1.1	1.2	0.0	1.0	3.0	302
Avg. marginal price: 0–3 kgal (\$/kgal)	5.3	4.0	0.0	4.8	11.2	350
Avg. marginal price: 3–5 kgal (\$/kgal)	8.8	3.8	4.3	8.5	13.7	350
Avg. marginal price: 5–10 kgal (\$/kgal)	9.1	4.0	4.6	8.7	14.4	350
Avg. marginal price: 10–15 kgal (\$/kgal)	9.4	4.4	4.6	8.9	14.9	350
Avg. total bill: 5 kgal (\$/kgal)	12.3	4.4	7.3	11.7	18.3	350
Avg. total cost (\$/kgal)	16.5	10.6	6.4	13.0	34.5	350
Median household income (\$ 1,000s)	38.3	13.8	24.0	35.9	53.4	350
Median home value (\$ 1,000s)	144.9	107.3	76.3	116.9	222.2	350
Pct. owner occupied	62.9	12.8	48.2	62.4	79.6	350
Prop. White (%)	67.7	21.3	36.4	72.2	93.5	350
Prop. Black (%)	24.7	20.2	1.4	20.2	54.3	350
Prop. Hispanic (%)	7.5	7.9	0.7	5.1	16.8	350
No. weeks with severe drought, 2010-2019	15.6	10.6	5.5	13.0	30.0	350

Notes: All observations are at the utility level for first year present in sample. For Georgia, data begin in 2008; for North Carolina, data begin in 2010. Number of utilities increases over time. All dollar values presented are real and deflated to 2019 dollars. Annual number of customers served is linearly interpolated and extrapolated within a utility if values are missing.

Table D.2: Marginal cost regressions using trans-log-like cost function approach

	Water source					
	(1) Constant	(2) Quadratic	(3) Terciles	(4) Surface	(5) Ground	(6) Purchased
$\ln(Q)$	-0.64*** (0.24)	0.18 (0.13)	0.35** (0.14)	-1.11*** (0.24)	0.36** (0.17)	0.15 (0.17)
$\ln(Q) \times 1[\text{Wat. only}]$	-0.06 (0.04)	-0.20*** (0.03)	-0.21*** (0.05)	-0.04 (0.07)	-0.22*** (0.03)	-0.21*** (0.04)
$\frac{1}{2} (\ln(Q))^2$	0.11*** (0.02)	0.05*** (0.01)	0.04*** (0.01)	0.14*** (0.02)	0.02 (0.02)	0.04*** (0.01)
$\frac{1}{2} (\ln(Q))^2 \times 1[\text{Wat. only}]$	0.00 (0.01)	0.02*** (0.01)	0.03*** (0.01)	0.00 (0.01)	0.03*** (0.01)	0.03*** (0.01)
Observations	1594	3192	1514	1551	3097	1450
No. Utilities	170	383	176	167	375	170
R-squared	0.94	0.91	0.93	0.95	0.92	0.95
State FE?	Y	Y	Y	Y	Y	Y
HUC6-by-Year FE?	Y	Y	Y	–	–	–
HUC6-by-Year-by-Size FE?	–	–	–	Y	Y	Y

Notes: The dependent variable is logged annual operating expenditures at the utility level. “1[Wat. only]” indicates utilities that only provide water service. The first column estimates a constant marginal cost across all utilities. Regressions are estimated separately for each water source (Surface, Groundwater, and Purchased). HUC6-by-Year-by-Size fixed effects refers to six-digit hydrological region-by-year-by-utility size, where utility size captures three categories by number of customers: 0–1,000, 1,000–5,000, and >5,000. Standard errors are clustered at the utility level.

Table D.3: Revenue elasticity estimates without fixed service fee

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$\widetilde{AMP}_{iht}$	0.13*** (0.02)			0.42*** (0.12)	0.45*** (0.08)		
$\widetilde{AP}_{iht}$			0.30*** (0.02)			0.60*** (0.16)	0.26*** (0.04)
$\widetilde{AMP}_{iht} : 0 - 3kgal$		-0.01 (0.02)					
$\widetilde{AMP}_{iht} : 3 - 5kgal$		0.11** (0.05)					
$\widetilde{AMP}_{iht} : 5 - 10kgal$		0.23** (0.11)					
$\widetilde{AMP}_{iht} : 10 - 15kgal$		0.04 (0.12)					
Observations	5085	5085	5085	5085	5085	5085	5085
No. Utilities	685	685	685	685	685	685	685
ϵ_p	-0.87	-0.77	-0.70	-0.58	-0.55	-0.40	-0.74
SE(ϵ_p)	0.02	0.11	0.02	0.12	0.08	0.16	0.04
IV: HUC6 $\widetilde{AMP}_t (t-1)$	—	—	—	Y	—	Y	—
IV: HUC6 mean rates $(t-1)$	—	—	—	—	Y	—	Y
Cragg-Donald Wald F stat	—	—	—	66.9	39.0	64.2	137.8

Notes: The dependent variable is the inverse hyperbolic sine of annual revenue per customer. All regressions include utility and HUC6×year fixed effects. Standard errors are clustered at the utility level. $\widetilde{AMP}_{iht}$ is the inverse hyperbolic sine of the average marginal price for each utility-year. In Column (2), I include each average marginal price in each consumption tier. $\widetilde{AP}_{iht}$ is the average volumetric price for each utility-year. In Columns (4) and (6), $\widetilde{AMP}_{iht}$ or $\widetilde{AP}_{iht}$ is instrumented with the lagged, simulated marginal price of the mean rate structure from all other utilities in the same HUC6 region. In Columns (5) and (7), $\widetilde{AMP}_{iht}$ or $\widetilde{AP}_{iht}$ is instrumented with the mean marginal prices in each price tier for all utilities in the same in HUC6 region. The inverse hyperbolic sine is applied to all base fees and price variables.

Table D.4: Revenue elasticity estimates in natural logs

Panel A: Revenue elasticities							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$\ln(AMP_{iht})$	0.23*** (0.02)			0.74** (0.30)	0.50*** (0.08)		
$\ln(AP_{iht})$			0.27*** (0.03)			0.66*** (0.22)	0.32*** (0.05)
$\ln(AMP_{iht} : 0 - 3kgal)$		-0.02 (0.03)					
$\ln(AMP_{iht} : 3 - 5kgal)$		0.12 (0.10)					
$\ln(AMP_{iht} : 5 - 10kgal)$		0.28** (0.11)					
$\ln(AMP_{iht} : 10 - 15kgal)$		-0.00 (0.12)					
Observations	4624	4254	4624	4624	4624	4624	4624
No. Utilities	639	590	639	639	639	639	639
ϵ_p	-0.77	-0.72	-0.73	-0.26	-0.50	-0.34	-0.68
SE(ϵ_p)	0.02	0.11	0.03	0.30	0.08	0.22	0.05
IV: HUC6 $\ln(AMP)$ ($t - 1$)	—	—	—	Y	—	Y	—
IV: HUC6 mean rates ($t - 1$)	—	—	—	—	Y	—	Y
Cragg-Donald Wald F stat	—	—	—	60.8	68.4	64.4	87.5
Panel B: Revenue elasticities including fixed fees							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$\ln(\text{Fixed fee}_{iht})$	0.20*** (0.03)	0.14*** (0.03)	0.23*** (0.03)	0.27*** (0.09)	0.31*** (0.09)	0.37*** (0.08)	0.40*** (0.07)
$\ln(AMP_{iht})$	0.22*** (0.02)			0.90** (0.40)	0.45*** (0.08)		
$\ln(AP_{iht})$			0.28*** (0.03)			0.85** (0.35)	0.34*** (0.05)
$\ln(AMP_{iht} : 0 - 3kgal)$		0.01 (0.03)					
$\ln(AMP_{iht} : 3 - 5kgal)$		0.07 (0.10)					
$\ln(AMP_{iht} : 5 - 10kgal)$		0.24** (0.10)					
$\ln(AMP_{iht} : 10 - 15kgal)$		-0.00 (0.12)					
Observations	4455	4094	4455	4455	4455	4455	4455
No. Utilities	631	582	631	631	631	631	631
ϵ_p	-0.78	-0.76	-0.72	-0.10	-0.55	-0.15	-0.66
SE(ϵ_p)	0.02	0.10	0.03	0.40	0.08	0.35	0.05
IV: HUC6 $\ln(AMP)$, $t - 1$	—	—	—	Y	—	Y	—
IV: HUC6 mean rates, $t - 1$	—	—	—	—	Y	—	Y
Cragg-Donald Wald F stat	—	—	—	22.5	49.5	20.7	70.2

Notes: The dependent variable is the log of annual revenue per customer. All regressions include utility and HUC6 $\times$ year fixed effects. Standard errors are clustered at the utility level. $\ln(AMP_{iht})$ is the log of the average marginal price for each utility-year. In Column (2), I include each average marginal price in each consumption tier. $\ln(AP_{iht})$ is the log of average volumetric price for each utility-year. In Columns (4) and (6), $\ln(AMP_{iht})$ or $\ln(AP_{iht})$ is instrumented with the lagged, simulated marginal price of the mean rate structure from all other utilities in the same HUC6 region. In Columns (5) and (7), $\ln(AMP_{iht})$ or $\ln(AP_{iht})$ is instrumented with the mean marginal prices in each price tier for all utilities in the same in HUC6 region. In Columns (4)–(7), the base service fee is instrumented with the mean base fee of all other utilities in the same HUC6 region. A log transformation is applied to all fixed service fees and price variables.

Table D.5: Price elasticity estimates with quantities

Panel A: Price elasticities							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$\widetilde{AMP}_{iht}$	0.26*** (0.03)			1.78*** (0.48)	-0.30** (0.12)		
$\widetilde{AP}_{iht}$			0.37*** (0.04)			2.51*** (0.74)	-0.14 (0.09)
$\widetilde{AMP}_{iht} : 0 - 3kgal$		-0.02 (0.02)					
$\widetilde{AMP}_{iht} : 3 - 5kgal$		-0.11 (0.11)					
$\widetilde{AMP}_{iht} : 5 - 10kgal$		-0.31* (0.16)					
$\widetilde{AMP}_{iht} : 10 - 15kgal$		-0.10 (0.14)					
Observations	6333	6333	6333	5085	5085	5085	5085
No. Utilities	719	719	719	685	685	685	685
IV: HUC6 $\widetilde{AMP}$, $(t - 1)$	-	-	-	Y	-	Y	-
IV: HUC6 mean rates, $t - 1$	-	-	-	-	Y	-	Y
Cragg-Donald Wald F stat	-	-	-	66.89	39.00	64.16	137.81
Panel B: Price elasticities including fixed fees							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Fixed fee $_{iht}$	-0.16*** (0.05)	-0.16*** (0.04)	-0.13*** (0.05)	-0.19 (0.15)	-0.22** (0.09)	0.03 (0.19)	-0.23** (0.09)
$\widetilde{AMP}_{iht}$	0.26*** (0.03)			1.71*** (0.46)	-0.29*** (0.11)		
$\widetilde{AP}_{iht}$			0.36*** (0.04)			2.53*** (0.77)	-0.18** (0.09)
$\widetilde{AMP}_{iht} : 0 - 3kgal$		-0.06** (0.03)					
$\widetilde{AMP}_{iht} : 3 - 5kgal$		-0.08 (0.10)					
$\widetilde{AMP}_{iht} : 5 - 10kgal$		-0.27* (0.15)					
$\widetilde{AMP}_{iht} : 10 - 15kgal$		-0.09 (0.14)					
Observations	6333	6333	6333	5085	5085	5085	5085
No. Utilities	719	719	719	685	685	685	685
IV: HUC6 $\widetilde{AMP}$, $(t - 1)$	-	-	-	Y	-	Y	-
IV: HUC6 mean rates, $t - 1$	-	-	-	-	Y	-	Y
Cragg-Donald Wald F stat	-	-	-	31.14	31.45	27.32	103.06

Notes: The dependent variable is the inverse hyperbolic sine of annual quantity consumed per customer. All regressions include utility and HUC6 $\times$ year fixed effects. Standard errors are clustered at the utility level. $\widetilde{AMP}_{iht}$ is the inverse hyperbolic sine of the average marginal price for each utility-year. In Column (2), I include each average marginal price in each consumption tier. $\widetilde{AP}_{iht}$ is the average volumetric price for each utility-year. In Columns (4) and (6), $\widetilde{AMP}_{iht}$ or $\widetilde{AP}_{iht}$ is instrumented with the lagged, simulated marginal price of the mean rate structure from all other utilities in the same HUC6 region. In Columns (5) and (7), $\widetilde{AMP}_{iht}$ or $\widetilde{AP}_{iht}$ is instrumented with the mean marginal prices in each price tier for all utilities in the same in HUC6 region. In Columns (4)–(7), the base service fee is instrumented with the mean base fee of all other utilities in the same HUC6 region. The inverse hyperbolic sine is applied to all base fees and price variables.

Table D.6: Welfare effects of switching to marginal-cost pricing - average price responsiveness

Panel A: Percentage change in quantity demanded ($\% \times 100$)					
	$\epsilon_p = -0.3$	$\epsilon_p = -0.4$	$\epsilon_p = -0.5$	$\epsilon_p = -0.6$	$\epsilon_p = -0.7$
IBR	-12.6	-16.7	-20.9	-25.1	-29.3
Uniform	4.1	5.5	6.8	8.2	9.6
Uniform + free	-67.5	-90.0	-112.5	-135.0	-157.5
Other	-14.5	-19.4	-24.2	-29.0	-33.9
Total	-23.1	-30.8	-38.5	-46.2	-53.9

Panel B: Annual deadweight loss from inefficient pricing (\$/customer)					
	$\epsilon_p = -0.3$	$\epsilon_p = -0.4$	$\epsilon_p = -0.5$	$\epsilon_p = -0.6$	$\epsilon_p = -0.7$
IBR	11.0	14.6	18.3	21.9	25.6
Uniform	3.9	5.2	6.5	7.8	9.1
Uniform + free	17.6	23.4	29.3	35.1	41.0
Other	14.0	18.7	23.3	28.0	32.7
Total	11.4	15.1	18.9	22.7	26.5

Panel C: Annual change in consumer surplus (\$/customer)					
	$\epsilon_p = -0.3$	$\epsilon_p = -0.4$	$\epsilon_p = -0.5$	$\epsilon_p = -0.6$	$\epsilon_p = -0.7$
IBR	-29.1	-28.9	-28.8	-28.7	-28.6
Uniform	13.0	13.0	13.0	13.1	13.1
Uniform + free	-155.5	-147.4	-142.7	-136.5	-121.1
Other	-32.6	-28.0	-23.3	-18.6	-14.0
Total	-61.6	-58.2	-55.5	-53.6	-51.6

Panel D: Annual change in municipal revenue (\$/customer)					
	$\epsilon_p = -0.3$	$\epsilon_p = -0.4$	$\epsilon_p = -0.5$	$\epsilon_p = -0.6$	$\epsilon_p = -0.7$
IBR	20.2	17.2	10.9	-4.0	-15.0
Uniform	-9.0	-7.8	-6.5	-5.2	-7.0
Uniform + free	101.7	75.8	42.0	12.0	-3.8
Other	18.6	9.3	4.5	-17.2	-34.5
Total	39.9	31.0	19.6	4.2	-9.9

Notes: Entries are annual statistics in 2015 for the representative customer in the median utility. The results include both combined water and wastewater utilities and water-only utilities. The representative customer is assumed to consume utility-specific mean quantity supplied each month of the year. For context, annual billed amounts under marginal-cost pricing (with $\epsilon = -0.5$) for the average utility are \$895, \$877, \$740, and \$723 for utilities that use IBRs or uniform, uniform + free, and other rates, respectively.

